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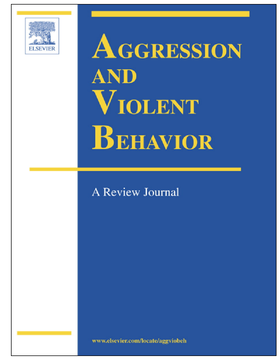

Cognitive functioning in criminal offenders: A multi-level meta-analysis

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JH, JvH, TZ and HG contributed to the conceptualization and methodology. The literature search was completed by JH. JH JvH and TZ did the article screening and full-text review. Data extraction and analysis were completed by JH, with guidance from GS. JH analyzed the risk of bias in the identified studies. JH wrote the first draft of the manuscript, and all authors commented on previous versions. All authors contributed substantially to the revision of the manuscript. All authors read and approved the final manuscript.

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Abstract

Cognitive functioning is increasingly recognized as an important factor in understanding offending behavior, yet findings across cognitive domains remain fragmented and inconsistent. This multi-level meta-analysis investigates the relationship between criminal offending and cognitive functioning. A search of PubMed, PSYCINFO, and Web of Science (January 2000-March 2025) yielded 101 studies ($N_{participants}$=12,789, $N_{effect\ sizes}$=1,757). Multi-level meta-analyses were conducted for each cognitive domain, comparing neuropsychological test performance between offenders and non-offending comparisons (non-offending clinical groups and general population groups). The meta-analyses confirmed weaker cognitive test performance in offenders compared to non-offending control groups. The average weighted effect size (g = 0.44) falls within the small to medium range. Significant differences were observed between offenders and comparisons across all DSM-5 neurocognitive domains: attention (g = 0.33); executive functions (g = 0.40); language (g = 0.53); learning/memory (g = 0.36); perceptual-motor functions (g = 0.24); and social cognition (g = 0.56). Tests differed in their ability to differentiate between offenders and comparison groups. Moderator analyses showed that observed differences in performance were largely absent when offenders were compared to clinical comparison groups, suggesting that difficulties may reflect general clinical dysfunction rather than offender-specific patterns. However, such difficulties could still play an indirect role in offending by reducing treatment responsivity or increasing the likelihood of dropout, both of which are known risk factors for reoffending.

Keywords: Neuropsychological Tests; Forensic Psychology; Criminal Behavior; Aggression; Meta-Analysis; Cognitive Functioning.

Introduction

Studies from various fields such as neuropsychology, developmental psychology, genetics, and neuropsychiatry have shown that cognitive functioning can play a significant role in the emergence and continuation of offending (Brower & Price, 2001; Jansen & Franse, 2024; Nauta-Jansen, 2022; Raine et al., 2005; Seguin et al., 2007). Offending refers to actions that violate laws, rules, or social norms, which can result in legal consequences. The current meta-analytic study examines how performance differences across six DSM-5 (Diagnostic and Statistical Manual of Mental Disorders; American Psychological Association, 2013) domains of cognitive functioning (attention, executive functions, language, learning/memory, perceptual-motor functions, and social cognition) are related to criminal offending (from now on: offending).

Difficulties in cognitive functioning are associated with various disorders that are reported to be more prevalent in offender populations compared to the general population. These include neurological conditions such as acquired brain injury (51% vs 12%; Farrer & Hedges, 2011; Frost et al., 2013; Williams et al., 2018), as well as psychiatric and developmental disorders such as intellectual disability (7-10% vs 1-3%; Harris, 2006; Hellenbach et al., 2017; Raitano Lee et al., 2016), attention-deficit/hyperactivity disorder (40% vs. 3.4%; Alderson et al., 2013; Fayyad et al., 2017; Ginsberg et al., 2010; Lukito et al., 2020), and externalizing disorders such as antisocial personality disorder (47% vs. 0.5-3.8%; De Brito et al., 2013; Fazel & Danesh, 2002; Lenzenweger et al., 2007; Trull et al., 2010). A significant portion of forensic psychiatric patients (9.5–35.7%) and correctional offenders (e.g., prisoners, parolees) (5.2–27.3%) show clinically significant deficits in all executive functioning components, substantially more than in the general population (2.5%; Shumlich et al., 2019). In addition, in a recent study among forensic outpatients, 21.1% performed below the MoCA cutoff (compared to 1.3% of the general population), suggesting mild cognitive impairments might be common in this population (Molleman et al., 2024).

The Risk-Need-Responsivity Model

To interpret these findings within a clinically relevant framework, the Risk-Need-Responsivity (RNR) model offers a useful lens. This model, developed by Bonta and Andrews (2017), identifies three key principles of effective offender rehabilitation: Risk, Need, and Responsivity. The Risk principle emphasizes aligning treatment intensity with the individual's risk of reoffending. The Need principle targets dynamic, changeable risk factors that contribute to offending behavior. The Responsivity principle focuses on tailoring interventions to individual characteristics that affect the success of treatment, such as learning style, motivation, or cognitive abilities. The following paragraphs explore how cognitive functioning relates to the core principles of the RNR model, with particular focus on its relevance to criminogenic needs and treatment responsivity.

Cognitive functioning: risk factor or responsivity factor?

Currently, most risk assessment instruments focus on observable behavior and symptoms, which are often interpreted through the lens of psychopathology. Assuming a maximum area-under-the-curve of 0.70 measured in most risk assessment instruments, it appears that the upper limit of predictive accuracy of recidivism has been reached (Ogonah et al., 2023). This ceiling suggests that existing models may overlook relevant individual characteristics such as neuropsychological factors that influence these behaviors and emotions. Although cognitive functioning is not typically included in current risk assessment tools, incorporating them as contextual information may help explain behavior and guide treatment planning. While research in this area is still limited, some studies suggest that adding neuropsychological and neurobiological information can improve the predictive validity of risk assessments by offering a more comprehensive, biopsychosocial perspective (Aharoni et al., 2013; Engel, 1977; Haarsma et al., 2020; Janes et al., 2024; Nauta-Jansen, 2022; Zijlmans et al., 2021).

However, instead of assuming cognitive functioning directly influences recidivism risk, it may be more accurate to understand them as indirect contributors to reoffending, i.e., responsivity factors. For example, cognitive difficulties in areas such as attention, memory, or language processing can make it harder for individuals to benefit from treatment. Executive functioning difficulties have also been associated with treatment non-compliance (Norman et al., 2023). However, some studies have found mixed results or no associations between cognitive functioning and treatment responses (Romero-Martínez et al., 2023; Van Der Sluys et al., 2020). These discrepancies may reflect methodological variation or different effects across offender subgroups or cognitive functions. In sum, while the extent to which cognitive functioning directly predicts recidivism remains uncertain, the influence on treatment responsivity is important to consider in research and clinical practice.

Cognitive functioning and the central eight

Cognitive functions may relate to the Need principle through their association with several of the Central Eight criminogenic needs (Cheng et al., 2019). These include antisocial behavior, antisocial attitudes, antisocial peers, poor family or marital relationships, poor school or work performance, substance abuse, and a lack of prosocial leisure activities (Bonta & Andrews, 2017). Difficulties in inhibition, planning, and cognitive flexibility have been associated with antisocial behavior and antisocial cognitions (Jansen & Franse, 2024; Ogilvie et al., 2011). These cognitive difficulties may help explain why some individuals struggle with behavioral regulation and social norm adherence, making them relevant to the Need principle. Similarly, impulsivity is connected to s*ubstance abuse*, another key risk factor in the Central Eight (Verdejo-Garcia & Albein-Urios, 2021).

In addition to these individual-level factors, cognitive functioning may also influence broader societal or contextual functioning, such as performance in work, school, or relationships. Attention and executive functions like planning, working memory, and cognitive flexibility contribute to success in these areas (Cortés Pascual et al., 2019). Difficulties in cognitive flexibility can result in rigid thinking patterns, hindering the ability to adapt thoughts (*antisocial cognitions*) or behaviors (*history of antisocial behavior*) based on changing social norms or consequences (Diamond, 2013). These associations between attention/executive functions and offending behavior have been supported by empirical studies. Meta-analyses comparing the attention and executive function task performance of antisocial individuals to control groups have shown a small to medium effect size (Jansen & Franse, 2024; Ogilvie et al., 2011). Although the term "antisocial" is broader than our study's specific focus on offenders (not all individuals exhibiting antisocial traits have committed crimes), these findings suggested meaningful average group difference in cognitive functioning. Ogilvie et al. (2011) separately analyzed various operationalizations of antisocial behavior of which the legal operationalization 'criminality' produced the largest effect size (d = 0.62).

Furthermore, difficulties with Theory of Mind (the ability to understand and attribute mental states to oneself and others; Klin, 2000) and emotion recognition, facets of social cognition, intersect with offending through mechanisms such as Social Information Processing (SIP; Crick & Dodge, 1994). This model involves the interpretation of social cues, which is more often distorted in offenders. Social-cognitive difficulties can lead to hostile attributions and reactive aggression, core components in the etiology of offending behavior (Smeijers, 2023). Additionally, the Central Eight factors *family/marital relationships* and *associations with antisocial others* may be influenced by social cognitive functioning. The link between social cognition and offending behavior has been confirmed by empirical studies. A meta-analysis covering 20 studies comparing antisocial and comparison groups found difficulties in recognizing facial expressions of fear, sadness, and surprise in antisocial samples (Marsh & Blair, 2008). No significant group differences were found for happiness, anger, and disgust. Results from research on Theory of Mind are mixed, but suggest potential difficulties in understanding particular mental states of others among offenders (Karoğlu et al., 2022).

Additionally, there is some evidence that offenders experience more language difficulties than in non-offending controls (Anderson et al., 2016; Chow et al., 2022; Cohen et al., 2003). Such difficulties may affect communication and contribute to challenges in social functioning. Learning disabilities, including difficulties in reading, writing, and processing information, are disproportionately represented within the offender population (Hellenbach et al., 2017; Muñoz García-Largo et al., 2020), and may be relevant to how individuals respond to feedback or adjust behavior in social contexts. Taken together, cognitive difficulties not only shape individual-level criminogenic needs, but also contribute to broader social and contextual difficulties.

Previous meta-analyses

A deeper understanding of how cognitive functioning can be understood from the RNR-framework offers a more nuanced approach to the rehabilitation of offenders, aligning interventions with individual cognitive profiles to optimize therapeutic impact and reduce offending behavior. Therefore, the first goal of this meta-analysis is to study which cognitive domains are related to offending. In doing so, we take a broader approach than prior meta-analyses that focused on executive functions (Jansen & Franse, 2024; Morgan & Lilienfeld, 2000; Ogilvie et al., 2011). Previous meta-analyses examined antisocial behavior, while we focus exclusively on offenders. To our knowledge, this is the first multi-level meta-analysis examining all cognitive domains in offenders compared to non-offending comparison groups.

To measure cognitive functioning, studies use a range of neuropsychological tests, which vary in how strongly they differentiate between offenders and non-offending controls

(Jansen & Franse, 2024; Morgan & Lilienfeld, 2000; Ogilvie et al., 2011). These discrepancies may reflect variations in underlying impairments, differences in test sensitivity, or both. Consequently, the second objective of this meta-analysis is to explore which tests show the largest performance differences between offenders and non-offending controls.

Current study

In sum, the findings of previous research suggest a relationship between offending behavior and cognitive domains, mainly for executive functions. To our knowledge, this is the first multi-level meta-analysis focusing specifically on offenders, whereas prior studies focused on broader antisocial samples. The primary aim of this multi-level meta-analysis is to summarize the current state of research on differences between offenders and non-offending comparison groups. The study focuses on the six neurocognitive domains as defined in the DSM-5: attention, executive functions, language, learning/memory, perceptual-motor functions, and social cognition. The second aim is to identify which neuropsychological tests and domains show the largest performance differences between offenders and non-offenders. In addition to the widely used RNR-model, we also adopt a biopsychosocial perspective to interpret our findings, considering biological, psychological, and social contributors to cognitive functioning in offender populations. Although our meta-analysis does not directly examine recidivism outcomes, the findings may help inform future work on the role of cognitive functioning in treatment engagement and long-term behavioral change.

Method

Search Strategy

We conducted a search in the databases PubMed, PSYCINFO and Web of Science covering the period January 2000 to March 2025. The year 2000 was selected as the starting point of the search, as forensic neuropsychology has grown substantially as a subspecialty since 2000, accompanied by growing standardization in assessment methods (Sweet et al., 2002;

Sweet et al., 2022; Sweet, 2024). In addition, earlier studies often relied on outdated versions of neuropsychological tests or did not align with the six cognitive domains examined in this study. Databases were searched for three key concepts: a) neuropsychological, comprising the six DSM-5 neurocognitive domains; b) assessment; and c) offenders. The DSM-5 domains were chosen because they provide a standardized and clinically relevant framework that aligns with clinical practice. We used the Boolean logic terms "OR" to search for synonyms and connected all three search terms using "AND". The complete search terms can be found in the supplementary materials. Additionally, we scrutinized the reference lists of relevant reviews to identify additional papers that could meet our eligibility criteria. Following the removal of duplicates, this search yielded 10.012 potentially relevant titles. We used the free, web-based application Rayyan (http://rayyan.qcri.org) for the screening of titles and abstracts (Ouzzani et al., 2016).

Eligibility Criteria

Studies had to meet the following inclusion criteria: a) written in English or Dutch; b) using a neuropsychological test or task to measure at least one of the DSM-5 cognitive domains (American Psychological Association, 2013); c) containing a comparison between an offender sample and a non-offending comparison sample. "Offenders" was defined as people who committed a criminal offense, were incarcerated for a criminal offense, and/or in forensic treatment. This focus excludes individuals exhibiting antisocial behavior without legal consequences, as our aim was to include populations relevant to forensic assessment and treatment. Studies were excluded if they met any of the following criteria: a) participants < 18 years old; b) studies that did not report original findings; c) studies that only used questionnaires or brain imaging; d) traffic violations and driving under influence (DUI); because in many countries, these are categorized as traffic or administrative violations and not as criminal offenses.

A total of 10,012 titles were screened by the first author, which led to the inclusion of 1,151 potentially relevant abstracts. A subsample of 750 abstracts were independently assessed by the last author using a manual to calculate interrater agreement, resulting in Cohen's κ = 0.80 (90.4% agreement). In the full text round, the second and last author independently assessed 100 full texts, resulting in κ = 0.66 (82.8% agreement). Later, another 30 full texts were independently assessed by the last author – among which the conflicts from the first round –, resulting in κ = 0.92 (96.7% agreement). In all rounds, any discrepancies were discussed until consensus was reached.

[INSERT FIGURE 1 HERE]

Data Extraction

The following study characteristics were extracted and entered in an SPSS datafile: (1) authors, (2) publication year, (3) location (country) of the study, (4) conflict of interest reported by the authors, (5) total number of participants in the study.

The following sample characteristics were extracted for each subsample separately: (1) sample size, (2) age (M, SD), (3) proportion of males, (4) years of education[1] (M, SD), (5) proportion from an ethnic minority group, (6) psychiatric diagnosis, (7) medication and substance use, (8) exclusion of traumatic brain injury and/or intellectual disabilities in the sample, (9) whether the samples were matched for education years/level and/or intelligence. For offender sample, we additionally extracted (1) type of forensic setting (prison, inpatient, outpatient, probation, assessment center, other), (2) offender type (general violence, property, domestic violence, sexual offense, other). We separated general violence and domestic violence as previous work suggests that different forms of violence might be underpinned by distinct neuropsychological mechanisms (Cruz et al., 2020; Humenik et al., 2020). For

[1] In case only the level of education was known, this was converted into years using the standard number of years of that education level per country.

comparison samples, the type of comparison sample (clinical: individuals with a psychiatric diagnosis vs. general: individuals without a psychiatric diagnosis) was extracted.

The following test outcomes were extracted: (1) the name of the (sub)test and the specific outcome measure (e.g., number of errors, reaction time), (2) test scores (M and SD), (3) if the test scores were unavailable: the reported effect size or statistic that was later calculated into *g* (η^2, χ^2, U, t) (4) number of participants in both samples that completed the test, (5) the cognitive domain(s) the tests aim to measure. As some tests contained different outcomes that measure different cognitive domains, these outcomes were assigned to domains separately. In addition, for the analyses on test-level, some tests were grouped together with their equivalents to facilitate interpretation. For example, different versions of digit span tests were analyzed together. The classification of tests into cognitive domains was based on content of test manuals and neuropsychology textbooks, and was checked by two senior neuropsychologists (TZ, HG) (see Supplementary Table S1). The merging of tests with their equivalents was checked by the same neuropsychologists. Any discrepancies were discussed until consensus was reached.

Quality assessment

To assess the quality of the included studies, an adapted version of the Newcastle-Ottawa Scale for cohort and cross-sectional studies (NOS; Wells et al., n.d.; adaptation by Jansen & Franse, 2024). The items ‘Ascertainment of exposure’ and ‘Same method of ascertainment for cases and controls’ were excluded, because they are not applicable to our study. Thus, the NOS includes seven items, with a star assigned each time criteria are met (A), and half a star assigned each time the criteria are partially met (B).

Data Synthesis and Analysis

Data were analyzed using Statistical Package for the Social Sciences (SPSS; IBM Corp, 2021) and “R” (R Core Team, 2021). Hedges g was calculated from the means, standard deviations and sample sizes if provided (Lakens, 2013; formula 4), or converted from other metrics such as partial eta squared. Positive values indicated worse performance in the offender sample. Effect sizes were interpreted using Cohen's (2009) guidelines: (0.2 = small, 0.5 = medium, 0.8 = large. We identified potential outliers via z-values (cutoff ± 3.29), verified their accuracy, and repeated analyses with substituted g-values the outliers to compare results.

Multi-level approach to meta-analysis

To account for dependent effect sizes within studies, we used three-level meta-analysis models (Cheung, 2014), modeling sampling variance (level 1), within-study variance (level 2), and between-study variance (level 3) (Cheung, 2014; Hox et al., 2018; Van den Noortgate et al., 2013). We used the function “rma.mv” of the metafor package (Viechtbauer, 2010, 2022) in the R environment (R Core Team, 2015). The R script and protocol were based on procedures outlined in methodological papers, modeling the three sources of variance (Assink & Wibbelink, 2016; Van den Noortgate et al., 2013, 2015). The approach of using the *t*-distribution for testing individual regression coefficients and calculating confidence intervals in meta-analytic models was employed in this study. This approach, as suggested by Knapp and Hartung (2003), accounts for the uncertainty in the residual variance, resulting in a more precise estimate of standard errors and a reduction in type-I errors. Various plots were made to facilitate interpretation of the results, using the R package *ggplot2* (Wickham et al., 2023).

Moderator analyses

Moderator analyses were explorative, and the choice of moderators was based on a) moderators that are frequently analyzed in meta-analyses (e.g., publication year) and b) variables that are known to be related to cognitive functioning (e.g., brain injury, age). Moderator analyses were performed for 17 study-, sample- and test characteristics. Study characteristics were publication year and study location. Sample characteristics were: age, percentage male, percentage from a minority group, groups matched for intelligence and/or education, groups matched for psychiatric diagnosis, exclusion of traumatic brain injury, exclusion of intellectual disabilities, type of offender group (general violence, domestic violence, sexual, property), offender group setting (prison, inpatient, outpatient), offender group diagnosis (none, neurodevelopmental disorders, schizophrenia spectrum and other psychotic disorders, personality disorders – according to the DSM-5 categorization). Test characteristics were the primary cognitive construct (domain) which the test was designed to measure. Mixed samples (e.g., consisting of both prison and inpatient participants) were excluded from moderator analyses. There was insufficient data for the moderators substance use, medication use, socioeconomic status (SES), and reported conflict of interest by the authors, precluding an examination of these possible moderators.

Moderator analyses were not performed if there were a) less than three effect sizes (continuous moderators); b) less than three studies per category (binary moderators); c) less than three effect sizes per category (categorical moderators). If necessary and appropriate, categories were merged to ensure the minimum number of effect sizes needed. Dichotomous dummy variables were created for all categorical variables, and continuous variables were centered around their mean (Tabachnick et al., 2019). In multilevel regression analyses, the intercept serves as the reference category, while the dummy variables test the degree to which the other categories differ from this reference category. To correct for multiple testing, the Benjamini-Hochberg False Discovery Rate (FDR) method was applied (Benjamini & Hochberg, 1995).

Heterogeneity

Heterogeneity across studies was reported using Q (25% = low, 50% = moderate, 75% = high Higgins, 2003; Higgins & Thompson, 2002). In addition, σ^2 statistics at level 2 and level 3 are reported as recommended for multi-level meta-analysis (Assink & Wibbelink, 2016).

Publication bias

To reduce the risk of publication bias, we contacted authors and were able to included two unpublished datasets. Additionally, we applied three statistical techniques to assess potential bias: 1) an adapted version of Egger's test for funnel plot asymmetry, 2) a multilevel version of the funnel plot, and 3) the trim-and-fill method using cutoff-based criteria (Duval & Tweedie, 2000; Fernández-Castilla et al., 2021). For details on these procedures, see (Fernández-Castilla, Declercq, et al., 2020; Fernández-Castilla, Jamshidi, et al., 2020; Roest et al., 2022).

Transparency and Openness

This meta-analysis was guided by the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA; Page et al., 2021; see Supplementary tables S11 and S12), and the Journal Article Reporting Standards (JARS) for Quantitative Meta-Analysis (American Psychological Association, n.d.). The study was pre-registered in the prospective register of systematic review [Masked for Blind Review]. At the time of pre-registration, authors were planning on conducting a systematic review only. To improve the robustness of results, it was later decided to perform a multi-level meta-analysis. Therefore, we did preregister our search terms and inclusion criteria, but not the analyses.

Results

Literature Search

A flowchart of our search strategy and screening process is shown in Figure 1. Studies included in the meta-analysis are listed in supplementary Table S2. Full-text assessment of articles resulted in 103 papers (k) meeting inclusion criteria for this multi-level meta-analysis. When multiple papers reported on the exact same sample, they were coded as a single study. After this consolidation and the addition of one unpublished dataset, the final meta-analysis included 101 studies with a total of 1,757 effect sizes and 12,789 participants.

Study Characteristics

Most studies were conducted in Western Europe (*k* = 41; 40.6%), 29 (28.7%) studies were carried out in other European countries, and 3 were conducted in multiple countries across Europe (3.0%)[2]. Another 18 studies took place in North America (17.8%), 7 studies in Oceania/Asia (7.0%), and the remaining 2 studies took place in South America (2.0%).

The total sample size across all studies was 12,798, with individual study sample sizes ranging from 14 to 1,210 participants. Approximately half of the studies compared one offender sample to one comparison sample (*k* = 54). The other 47 studies included more than one comparison (e.g., two offender samples vs. one comparison sample, or vice versa). In total, 142 different offender samples and 122 comparison samples were included.

Out of the 122 comparison samples, 30 (24.6%) were clinical comparisons (CC) with a psychiatric diagnosis but who did not commit any offense, 92 (75.4%) were composed of individuals without a known psychiatric diagnosis (general comparisons: GC). Of the 142 offender samples, 32 (22.5%) were violent offenders, 22 (15.5%) committed sexual offenses, 16 (11.3%) were domestic violence offenders, 5 (3.5%) committed property crimes, 31 (21.8%) samples were mixed, and for 36 (25.4%) samples, the crime type was not described. The offender samples were situated in different settings such as prisons (54; 38.0%), inpatient facilities (34; 23.9%), outpatient facilities (19; 13.4%), probation (6; 4.2%), assessment centers (2; 1.4%), a combination of one or more of the above (22; 15.5%), and for 2 samples (1.4%), the setting was unknown. Offender participants had a mean age of 37.3 years old (SD = 6.2) and comparisons 35.7 years old (SD = 7.9). Regarding the sex distribution across samples, 114 of the 142 (80.3%) offender samples were male-only, compared to 87 of the 122 (71.3%)

[2] European countries were divided into regions using the European Vocabularies, see: https://op.europa.eu/en/web/eu-vocabularies/concept-scheme/-/resource?uri=http://eurovoc.europa.eu/100277

comparison samples. The mean (SD) years of education for offenders was 10.5 (2.0) and 12.3 (2.5) for the comparison samples.

Across all studies, a total of 188 different tests were used (counting battery subtests as separate tests). These outcomes were categorized into the six cognitive domains. For 87 test outcomes, it was impossible to categorize them into one of the domains as they measured multiple cognitive domains (see Supplementary Table S1). These outcomes were included in the overall meta-analysis, but excluded from domain-level analyses.

Methodological quality

Individual quality ratings according to the NOS (Wells et al., n.d.; adaptation by Jansen & Franse, 2024) can be found in the Supplementary material (Table S3). See Figure 2 for a summary of study quality. As the NOS utilizes stars as the scoring system, qualitative descriptors were utilized to categorize papers for ease of interpretation and explanation. Generally, study quality was fair to high. Thirty-three papers (32.7%) were considered high quality (6-7 stars), 58 (57.4%) fair quality (4-5.5 stars), ten (9.9%) were low quality (2-3.5 stars), and none were of unacceptable quality (0-1 stars). There was a risk of bias concerning the representativeness of the cases, suggesting a potential selection bias in the recruitment process at the forensic facility for all studies. This could mean, for example, that the sample might have predominantly included the most motivated offenders in the facility.

[INSERT FIGURE 2 HERE]

Overall comparison of offenders and comparisons

The mean overall estimate was significant (g = 0.44 [0.35, 0.52]), indicating a small to medium difference between offenders and comparison groups, which shows that overall, offenders performed worse on neuropsychological tests.

Neuropsychological Domains

Six different meta-analyses were carried out, one for each neurocognitive domain (see Table 1 and Figure 3, and supplementary Figures S1-S7 for forest plots). Significant differences were observed between offenders and comparisons across all neuropsychological domains: attention (g = 0.33 [0.23, 0.42]), executive functions (g = 0.40 [0.31, 0.48]), language (g = 0.53 [0.26, 0.80]), learning/memory (g = 0.36 [0.22, 0.50]), perceptual-motor functions (g = 0.24 [0.03, 0.45]), and social cognition (g = 0.56 [0.35, 0.78]). The three core executive functions were analyzed separately: cognitive flexibility (g = 0.51 [0.36, 0.66]), working memory (g = 0.44 [0.31, 0.48]), and inhibition (g = 0.26 [0.15, 0.37]). Cognitive domain was also investigated as a moderator to statistically compare the differences in effect size between the domains (see Table 2). With perceptual-motor functions as the reference category, there was only a significant difference with executive functions (g = 0.44 vs. 0.31, $p < .05$) and language (g = 0.61 vs. 0.31, $p < .05$)[3]. However, variations in the number of studies and effect sizes suggest that power issues may influence these findings.

[INSERT TABLE 1 HERE]

[INSERT FIGURE 3 HERE]

Sensitivity Analyses

There were 16 outliers in the overall meta-analysis[4]. Sensitivity analysis with corrected g-values yielded nearly identical results, indicating that the observed effects are not driven by extreme values.

[3] Effect sizes from the separate domain analyses may differ slightly from those in the moderator model, which estimates all domains together and adjusts for differences between studies.
[4] Outliers with z > 3.29: nine from Mariani et al. (2024), the faux pas test; one from Easton et al. (2008), the Iowa Gambling Task; one from Ostroysky et al. (2012), the Maze test; two from Castellino et al. (2012), the Thomas test; one from Cohen et al. (2010), Matching Familiar Figure Test (MFFT); and one from Spenser et al. (2015), the social stories questionnaire. Outliers with z < -3.29: one from Easton et al. (2008), the Trail Making Test.

Neuropsychological Tests

Next, performance on neuropsychological tests was examined (Supplementary Table S4). Only tests that were used in two or more studies with a minimum of three effect sizes were analyzed, and some tests that were equivalents were analyzed together (see table note). This left 49 tests or test categories to be analyzed.

Offenders performed significantly worse on 24 out of 49 neuropsychological tests than comparisons. Large (> 0.8), significant effects were found for semantic fluency tests (g = 1.08 [0.40, 1.77], p = .005, k = 6), Letter Number Sequencing (g = 1.09 [0.73, 1.46], k = 4), WAIS Vocabulary (g = 0.91 [0.12, 1.70], k = 5), and the CANTAB Spatial Span (g = 0.81 [0.15, 1.45], k = 4). In addition, nine tests produced significant medium effect sizes (between 0.5 and 0.8): The California Verbal Learning Test, CANTAB Rapid Visual Processing, CANTAB Attention Switching Task, Movie for the Assessment of Social Cognition (MASC), Wisconsin Card Sorting Test (WCST), Phonemic fluency tests, D2, WMS logical memory, WMS Spatial Span, and Reading the Mind in the Eyes (RTME).

Heterogeneity

Our meta-analytic approach identified significant heterogeneity (see Table 1). Only 8.7% of variance can be attributed to sampling error (level 1), while 49.3% and 41.9% were attributed to within-study (level 2) and between-study (level 3) differences, respectively (σ^2 = 0.19 and 0.16). According to Hunter and Schmidt's (1990) 75% rule, this indicates that most variance reflects real differences rather than sampling error, justifying moderator analyses.

Moderator analyses

To explore whether the addition of a moderator could explain (some of) the heterogeneity in effect sizes, moderator analyses were performed (see Table 2). If a moderator was significant[5] in the overall analysis, a domain-specific examination was conducted to assess variations in its moderating effects across domains (see Supplementary Tables S5-S10).

The following factors did not significantly moderate results are were therefore not examined at the domain level: cognitive domain, publication year, age, percentage male, percentage from a minority group, exclusion of brain damage, exclusion of intellectual disabilities, type of offender (general violence, domestic violence, sexual, or property crimes), and offender setting (prison, inpatient, or outpatient). Significant moderating effects are discussed below.

Study location significantly moderated the effect size (p = .003): studies conducted in non-Western Europe[6] reported larger effect sizes than studies from Western European countries. On domain-level, study location was significant for the domains attention (p = .012), executive functions (p = .037) and perceptual-motor functions (p < .001). Matching offender and comparison groups on intelligence and/or education also moderated results: matched studies showed smaller effects than unmatched ones (g = 0.36 vs. 0.87, p < .001), but this moderator was again only significant for the domain social cognition (g = 0.43 vs. 1.17, p < .001). Finally, the psychiatric status of the offender and comparison groups significantly moderated the effect size. When both groups did not have a psychiatric diagnosis, a moderate effect size was found (g = 0.57). A similar effect was observed when the offender group had a psychiatric diagnosis, but the comparison group did not (g = 0.56). However, when both groups included people with clinical diagnoses, the effect size was substantially reduced (g = 0.05).

Publication Bias

We used three methods to address publication bias (see Table 3). The funnel plot test indicated publication bias for learning/memory and perceptual-motor functions. Egger's test

[5] Using the Benjamini-Hochberg False Discovery Rate (FDR) method

[6] European countries were divided into regions using the European Vocabularies, see: https://op.europa.eu/en/web/eu-vocabularies/concept-scheme/-/resource?uri=http://eurovoc.europa.eu/100277

was significant for the overall meta-analysis, and for executive functions, learning/memory and perceptual-motor functions. The trim-and-fill method (Fernández-Castilla et al., 2021) suggested potential bias for the overall meta-analysis and for attention, learning/memory, and social cognition. These results suggest that the observed effects may be overestimated. For funnel plots see Figures 4a and 4b (for funnel plots per domain, see Supplementary Figures S8-S13).

[INSERT TABLE 3 HERE]

[INSERT FIGURE 4A AND 4B HERE]

Discussion

Summary findings

This meta-analysis provides empirical support for differences between offenders and non-offending comparisons in all domains of cognitive functioning (attention, executive functions, language, learning/memory, perceptual-motor functions, and social cognition). Based on 101 studies and 1,757 effect sizes, the results showed that offenders performed significantly worse in all cognitive domains compared to non-offending comparison groups. The overall weighted effect size was $g = 0.44$, signifying a small-to-medium effect size, which is comparable to previous meta-analyses of cognitive functioning and antisocial behavior ($d = 0.44$; Ogilvie et al., 2011 and $d = 0.42$; Jansen & Franse, 2024).

The second aim of this study was to identify which neuropsychological tests and cognitive domains show the largest performance differences between offenders and non-offending comparison groups. Offenders performed significantly worse on 24 of 49 tests, with largest effect sizes for Letter Number Sequencing (g = 1.09, k = 4), semantic fluency tests ($g = 1.08$, k = 6), and the WAIS Vocabulary subtest (g = 0.91, k = 5). Tests with the largest effect sizes may be particularly sensitive to cognitive difficulties offenders. However, the fact that only 24 of 49 tests showed significant differences also suggests that many used tests do not meaningfully distinguish these groups.

Theoretical explanations for differences between offenders and non-offending comparisons

Offending behavior often coexists with a range of neurobiological, psychological, and social issues. Cognitive dysfunction may contribute to various challenges in daily life, such as difficulties in decision-making, problem-solving, and maintaining social relationships. Consequently, these cognitive difficulties are likely related to a broader spectrum of problems, rather than being exclusively linked to offending behavior. We included only studies with non-offending control groups to examine how offenders, as a group, differ from the general population. Below, the findings from this meta-analysis are discussed from a biopsychosocial perspective.

First, on a neurobiological level, research has found reduced prefrontal structure and function in offenders (Yang & Raine, 2009). In addition, it is well known that acquired brain injury is more prevalent in offender populations than in the general population, with estimations of 51% versus 12% (Farrer & Hedges, 2011; Frost et al., 2013). However, acquired brain injury is only one of several biological influences on cognitive functioning. Neurodevelopmental factors such as genetic vulnerability, early life stress, and prenatal or perinatal complications, may disrupt brain maturation from an early age (Choy & Raine, 2024). These neurobiological differences could contribute to the poorer performance of offenders compared to the general population on neuropsychological tasks. In our meta-analysis, studies that excluded participants with brain injuries did not produce larger/smaller effect sizes than studies that did not exclude them. However, future studies should report on the presence of brain injury more consistently.

Second, psychological factors may play a role. Attention-deficit/hyperactivity disorder (ADHD), intellectual developmental disorder, substance-related disorders, and personality disorders are overrepresented in the offender population (De Vogel et al., 2022; Fayyad et al., 2017; Fazel & Danesh, 2002; Ginsberg et al., 2010; Harris, 2006; Hellenbach et al., 2017; Lenzenweger et al., 2007; Matheson et al., 2022; Muñoz García-Largo et al., 2020; Trull et al., 2010). These disorders are related to psychological factors such as self-regulation, impulse control, and emotional processing. Moderator analyses showed that when comparing offenders to the general population (i.e., no known psychiatric diagnosis), differences were larger than when comparing offenders to non-offenders with a psychiatric diagnosis. Effect sizes were close to zero when the offender and the non-offender groups both had a psychiatric diagnosis. Notably, 75% of the comparisons were made with general population groups, which may have inflated the overall effect size. This pattern may point toward the existence of a transdiagnostic cognitive (c-)factor, which is thought to underlie cognitive impairments across a range of psychiatric disorders (Abramovitch et al., 2021). However, a moderate effect size ($g = 0.57$) remained when both groups had no formally reported psychiatric diagnosis, indicating that the c-factor alone does not fully explain the group differences. Other factors such as adverse childhood experiences, socioeconomic status, or education, may also contribute to cognitive differences. Due to inconsistent reporting of these variables across studies, moderator analyses could not be conducted. It should also be noted that absence of a formal diagnosis does not guarantee absence of psychopathology. Therefore, these comparisons should be interpreted with caution.

Surprisingly, offender type (general violence, domestic violence, property or sexual), did not significantly moderate results. This contrasts with previous research suggesting that violent offenders cognitive performance differs from non-violent offenders (Cruz et al., 2020; Humenik et al., 2020). A possible explanation is the sample heterogeneity and broad or inconsistent definitions across studies included in the meta-analyses, which complicate comparisons.

Third, offenders, more often than individuals from the general population, face a number of social and environmental challenges, for example low socioeconomic status, exposure to violence, and low academic achievement (Baker et al., 2023; Kazeem, 2020; Savage et al., 2017). It is proposed that social and environmental challenges might exert an effect on antisocial behavior through the mediating role of neurobiological and neuropsychological limitations (Van Goozen et al., 2022). In our meta-analysis, we unfortunately did not have enough information on specific social and environmental challenges to study their potential moderating effects. Moreover, the intersection of multiple marginalized identities—such as being both an offender and having a mental disorder—can lead to compounded stigma and disadvantage, which may further interact with the aforementioned factors. Future research should take these intersecting influences into account to better understand how contextual and psychological factors jointly contribute to offending behavior.

Effect sizes were larger for social cognition (g = 0.56), language (g = 0.53) and cognitive flexibility (g = 0.51) than for other domains (e.g. perceptual motor, g = 0.22). These patterns likely reflect meaningful differences in cognitive functioning among offenders. However, variation in test suitability due to factors such as literacy, bilingualism, education, or familiarity with formal testing, may increase differences on certain tests. This highlights the need to critically evaluate which neuropsychological tests are valid and reliable for use in offender populations. The largest effect size was found for social cognition. Difficulties in this domain can directly impact interpersonal functioning, empathy, and treatment engagement. Despite growing interest, social cognition is not yet routinely assessed in many forensic settings. Notably, although language showed one of the largest effect sizes, relatively few studies (13) examined this domain. Given that language abilities influence communication, participation in therapy, and comprehension of therapeutic content, language represents an important responsivity factor. This pattern suggests that domains most directly related to interpersonal and communicative functioning may be particularly relevant in offender populations.

Implications of Findings

The biopsychosocial perspective helps us understand why offenders and non-offending comparisons differ in neuropsychological task performance by highlighting the interplay of neurobiological, psychological, and social factors. In contrast, the RNR- principles provide a framework for risk assessment and management. By integrating these perspectives, we can gain a comprehensive understanding of offender behavior and develop more effective strategies for rehabilitation and recidivism prevention.

First, outcomes of neuropsychological measures can potentially play an important role in improving recidivism predictions by pinpointing cognitive limitations, specifically in executive functions like decision-making and inhibition (Aharoni et al., 2013; Ross & Hoaken, 2011). Traditional risk assessment instruments seem to have reached a ceiling effect of a moderate area under the curve of 0.70 (Monahan & Skeem, 2014; Ogonah et al., 2023). So far, most risk assessment instruments focus on socio-behavioral factors such as an antisocial network and previous convictions (Bonta & Andrews, 2017), but do not include cognitive functioning as a predictors of (re-)offending behavior. Existing tools may assess constructs like impulsivity via self-report, whereas neuropsychological tasks offer a more objective and performance-based approach that does not rely on self-insight - often compromised in individuals with cognitive dysfunction (Steward & Kretzmer, 2022). Hence, incorporating specific neuropsychological tasks into current assessment procedures provides a more comprehensive understanding of offender's cognitive functioning, and could lead to more accurate risk predictions and to more responsive treatment, aligning with the RNR-principles.

Second, insight into cognitive limitations can guide risk management by facilitating the development of specific targeted interventions. It is important to highlight the importance of considering individual differences within offender populations when interpreting and applying these findings. For example, if an offender performs poorly on social cognition tasks, interventions could be aimed at improving social skills, perspective taking, and empathy. Likewise, if executive function problems are present, interventions should be aimed at enhancing impulse control, problem-solving, planning, and decision-making.

Third, a responsive treatment is one of the three pillars of offender rehabilitation (Bonta & Andrews, 2017). By being aware of the offenders' cognitive limitations, the clinician can adjust the treatment. For example, if working memory problems are identified, therapists may adjust by repeating information more often, using visual aids, or providing written summaries. These practical modifications can improve treatment engagement and retention, especially in forensic populations where dropout is common (Olver et al., 2011). This is particularly relevant given that individuals who are non-compliant with community-based supervision tend to report more executive functioning difficulties (Norman et al., 2023).

While the integration of neuropsychological measures into risk assessment and management shows promise, comprehensive assessment is unlikely to be feasible for all offenders due to time and resource constraints. A practical solution may be to adopt a stepped approach, using brief cognitive screeners or targeted screening questions to identify individuals who can benefit most from comprehensive neuropsychological assessment. Future studies should establish cut-offs for specific populations, as these may vary across groups. In addition, computerized tests may offer alternatives, as these can reduce clinician burden, although their validity and feasibility in forensic settings require further investigation.

Study Limitations

The results from this study should be observed considering a few limitations. First, due to inconsistencies across the studies included in reporting possible moderators, it was impossible to examine the potential influence of factors such as substance- and medication use. The prevalence of substance use problems or disorders is estimated at 30-50% in prisons and 18-69% in forensic mental health care settings (Butler et al., 2022; Fazel et al., 2017). We highlight the importance of examining potential moderators such as substance abuse, as executive functioning has been linked to substance use disorders (for a meta-analysis see Stephan et al., 2017).

Second, to overcome potential publication or selection bias analyses, we have attempted to obtain unpublished data from authors by contacting them via email multiple times. Since publication bias analyses methods have their limitations, we applied multiple methods to investigate potential publication bias (Assink & Wibbelink, 2016; Fernández-Castilla et al., 2021). As some of these methods gave indications of publication bias, publication bias cannot be ruled out.

A third limitation relates to the heterogeneity of the studies included. Although an overall effect size of $g = 0.44$ was found, there was substantial between-study variability in populations and cognitive tests that were used. While we tried to uncover these differences using moderator analysis, this suggests that other, unmeasured factors may explain part of the variance in effect sizes, which limits the generalizability of the results.

Recommendations for Future Research

There are several insights from neuropsychology which can be used in optimizing diagnosis and treatment of forensic patients. First, previous studies in the field of forensic neuropsychology have mainly focused on the domains attention and executive functions (see meta-analyses by Jansen & Franse, 2024; Morgan & Lilienfeld, 2000; Ogilvie et al., 2011).

The authors of a previous, similar meta-analysis by Ogilvie et al. (2011) suggest that a truer perspective might be that offending is related to a broader spectrum of neuropsychological limitations. Therefore, we took a broader approach by also including language, learning/memory, perceptual-motor functions, and social cognition. This idea is confirmed by the findings of the present study, emphasizing the importance of examining cognitive functions across multiple domains. Moreover, the neuropsychological domains are interconnected. To illustrate, attention is crucial for effective memory encoding and language processing (Chun & Turk-Browne, 2007; Turk-Browne et al., 2013), and executive functions can affect social cognition by guiding behavioral control and social decision-making (Gonzalez Aguilar, 2021; Von Hippel, 2007). In addition, memory is essential for retrieving grammar rules, semantic knowledge and vocabulary (Vijayalakshmi & Patchainayagi, 2020). These examples stress the importance of considering them as a network rather than as separated elements in future research (e.g., Guo et al., 2023).

Second, the results from this meta-analysis reveal a possible relation between cognitive functioning and offending. However, as the studies included were cross sectional in nature, it is impossible to determine whether cognitive difficulties increase the likelihood of offending behavior, or whether a criminal lifestyle causes cognitive difficulties, or both. For example, inhibition may increase the likelihood of engaging in high-risk situations that can lead to offending behavior. At the same time, involvement in criminal behavior can expose individuals to risk factors such as traumatic brain injury or substance use, that could in turn influence cognitive functions. Future studies should focus on understanding the pathways through which cognitive functioning and offending behavior might be related through prospective longitudinal study designs.

Third, the results from this meta-analysis emphasized the need for more uniformity in neuropsychological testing in the offender population. In total, 188 different neuropsychological tests were used across studies. This issue was also addressed in our previous Delphi study [Masked for Blind Review]. Future research could benefit from standardized assessment and methodologies by incorporating psychometrically validated tests, along with larger and more representative samples. This has been explored for patients with ADHD (Fuermaier et al., 2019), cancer (Noll et al., 2018) and schizophrenia (Nuechterlein et al., 2022).

Finally, more research is needed to examine the added value of neuropsychological factors for risk assessment and management, aligning with the RNR-principles. Future research should clarify how neuropsychological functioning contributes to risk assessment, intervention planning, and treatment responsivity. The integration of information on offenders' cognitive functioning with psychosocial information (e.g., risk assessment, questionnaires, etc.) marks an important step towards understanding and preventing offending behavior.

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The authors declare no competing interests.

Declaration of interests

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Availability of Data and Materials

Data is available upon request. Analytic code and supplementary tables and figures are available in the supplementary material. The supplementary material for this article can be found at [update to provide the DOI here]

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the author(s) used ChatGPT (GPT-5, OpenAI, 2025) to assist with English language editing and stylistic refinement. After using this tool/service, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the content of the published article.

Supplementary data

Supplementary material

References

* References marked with an asterisk Indicates included study.

Abramovitch, A., Short, T., & Schweiger, A. (2021). The C Factor: Cognitive dysfunction as a transdiagnostic dimension in psychopathology. Clinical Psychology Review, 86, 102007. https://doi.org/10.1016/j.cpr.2021.102007

Aharoni, E., Vincent, G. M., Harenski, C. L., Calhoun, V. D., Sinnott-Armstrong, W., Gazzaniga, M. S., & Kiehl, K. A. (2013). Neuroprediction of future rearrest. Proceedings of the National Academy of Sciences, 110(15), 6223–6228. https://doi.org/10.1073/pnas.1219302110

Alderson, R. M., Kasper, L. J., Hudec, K. L., & Patros, C. H. G. (2013). Attention-deficit/hyperactivity disorder (ADHD) and working memory in adults: A meta-analytic review. Neuropsychology, 27(3), 287–302. https://doi.org/10.1037/a0032371

American Psychological Association. (2013). Diagnostic and statistical manual of mental disorders—5th edition. American Psychological Association.

American Psychological Association. (n.d.). JARS–Quant | Table 9 Quantitative Meta-Analysis Article Reporting Standards. APA Style Journal Article Reporting Standards (JARS). Retrieved https://apastyle.apa.org/jars/quant-table-9.pdf

Anderson, S. A. S., Hawes, D. J., & Snow, P. C. (2016). Language impairments among youth offenders: A systematic review. Children and Youth Services Review, 65, 195–203. https://doi.org/10.1016/j.childyouth.2016.04.004

Assink, M., & Wibbelink, C. J. M. (2016). Fitting three-level meta-analytic models in R: A step-by-step tutorial. The Quantitative Methods for Psychology, 12(3), 154–174. https://doi.org/10.20982/tqmp.12.3.p154

*Baker, S., & Ireland, J. (2007). The link between dyslexic traits, executive functioning, impulsivity and social self-esteem among an offender and non-offender sample.

International Journal of Law and Psychiatry, 30(6), 492–503. https://doi.org/10.1016/j.ijlp.2007.09.010

Baker, S., Javakhishvili, M., & Spatz Widom, C. (2023). Childhood family and neighbourhood socio-economic status, psychopathy, and adult criminal behaviour. Legal and Criminological Psychology, 28, 106–121. https://doi.org/10.1111/lcrp.12228

*Baliousis, B., Duggan, C., McCarthy, Huband, N., & Völlm, B. (2019). Executive function, attention, and memory deficits in antisocial personality disorder and psychopathy. Psychiatry Research, 278, 151–161. https://doi.org/10.1016/j.psychres.2019.05.046

*Barkataki, I., Kumari, V., Das, M., Hill, M., Morris, R., O'Connell, P., Taylor, P., & Sharma, T. (2005). A neuropsychological investigation into violence and mental illness. Schizophrenia Research, 74(1), 1–13. https://doi.org/10.1016/j.schres.2004.08.001

*Barkataki, I., Kumari, V., Das, M., Sumich, A., Taylor, P., & Sharma, T. (2008). Neural correlates of deficient response inhibition in mentally disordered violent individuals. Behavioral Sciences & the Law, 26(1), 51–64. https://doi.org/10.1002/bsl.787

*Barlati, S., Nibbio, G., Stanga, V., Giovannoli, G., Calzavara-Pinton, I., Necchini, N., Lisoni, J., Deste, G., & Vita, A. (2023). Cognitive and clinical characteristics of offenders and non-offenders diagnosed with schizophrenia spectrum disorders: Results of the Recoviwel observational study. European Archives of Psychiatry and Clinical Neuroscience, 273(6), 1307–1316. https://doi.org/10.1007/s00406-022-01510-9

*Battaglia, A. M., Gicas, K. M., Mamak, M., & Goldberg, J. O. (2022). Mistakes in interpersonal perceptions: Social cognition in aggressive forensic psychiatry patients.

Criminal Behaviour and Mental Health, 32(1), 21–34. https://doi.org/10.1002/cbm.2228

*Becerra-García, J. A. (2014). Childhood abuse and neurocognitive performance in adulthood: A preliminary study of executive functioning in offenders. The Journal of Forensic Psychiatry & Psychology, 25(1), 113–119. https://doi.org/10.1080/14789949.2013.873070

*Becerra-Garcia, J. A., & Egan, V. (2014). Neurocognitive functioning and subtypes of child molesters: Poorer working memory differentiates incestuous from non-incestuous offenders. Psychiatry, Psychology and Law, 21(4), 585–590. https://psycnet.apa.org/doi/10.1080/13218719.2013.873974

Benjamini, Y., & Hochberg, Y. (1995). Controlling the False Discovery Rate: A Practical and Powerful Approach to Multiple Testing. Journal of the Royal Statistical Society Series B: Statistical Methodology, 57(1), 289–300. https://doi.org/10.1111/j.2517-6161.1995.tb02031.x

*Bergeron, T. K., & Valliant, P. M. (2001). Executive Function and Personality in Adolescent and Adult Offenders vs. Non-Offenders. Journal of Offender Rehabilitation, 33(3), 27–45. https://doi.org/10.1300/J076v33n03_02

*Bergvall, A. H., Nilsson, T., & Hansen, S. (2003). Exploring the link between character, personality disorder, and neuropsychological function. European Psychiatry, 18(7), 334–344. https://doi.org/10.1016/j.eurpsy.2003.03.008

*Beszterczey, S., Nestor, P. G., Shirai, A., & Harding, S. (2013). Neuropsychology of Decision Making and Psychopathy in High-Risk Ex-Offenders. Neuropsychology, 27(4), 491–497. https://doi.org/10.1037/a0033162

Bonta, J., & Andrews, D. A. (2017). The psychology of criminal conduct (6th edition). Routledge, Taylor & Francis Group.

*Brazil, I. A., Verkes, R. J., Brouns, B. H. J., Buitelaar, J. K., Bulten, B. H., & de Bruijn, E. R. A. (2012). Differentiating psychopathy from general antisociality using the P3 as a psychophysiological correlate of attentional allocation. PLoS ONE, 7(11), e50339. https://doi.org/10.1371/journal.pone.0050339

*Broom, I. (2011). The relationship between psychopathy and performance on a modified version of the Iowa Gambling Task in offender and undergraduate student samples [Doctor of Philosophy, Carleton University]. https://doi.org/10.22215/etd/2012-09628

Brower, M. C., & Price, B. H. (2001). Neuropsychiatry of frontal lobe dysfunction in violent and criminal behaviour: A critical review. Journal of Neurology, Neurosurgery & Psychiatry, 71(6), 720–726. https://doi.org/10.1136/jnnp.71.6.720

*Bulgari, V., Iozzino, L., Ferrari, C., Picchioni, M., Candini, V., De Francesco, A., Maggi, P., Segalini, B., & De Girolamo, G. (2017). Clinical and neuropsychological features of violence in schizophrenia: A prospective cohort study. Schizophrenia Research, 181, 124–130. https://doi.org/10.1016/j.schres.2016.10.016

Butler, A., Nicholls, T., Samji, H., Fabian, S., & Lavergne, M. R. (2022). Prevalence of Mental Health Needs, Substance Use, and Co-occurring Disorders Among People Admitted to Prison. Psychiatric Services, 73(7), 737–744. https://doi.org/10.1176/appi.ps.202000927

*Canessa, N., Iozzino, L., Andreose, S., Castelletti, L., Conte, G., Dvorak, A., Ferrari, C., Heitzman, J., Macis, A., Markiewicz, I., Mattavelli, G., Nicolò, G., Picchioni, M., Restuccia, G., Rivellini, G., Teti, F., & de Girolamo, G. (2023). RISK aversion in Italian forensic and non-forensic patients with schizophrenia spectrum disorders. PLoS ONE, 18(7), e0289152. https://doi.org/10.1371/journal.pone.0289152

*Castellino, N., Bosco, F. M., Marshall, W. L., Marshall, L. E., & Veglia, F. (2011). Mindreading abilities in sexual offenders: An analysis of theory of mind processes.

Consciousness and Cognition: An International Journal, 20(4), 1612–1624. https://doi.org/10.1016/j.concog.2011.08.011

Cheng, J., O'Connell, M. E., & Wormith, J. S. (2019). Bridging Neuropsychology and Forensic Psychology: Executive Function Overlaps With the Central Eight Risk and Need Factors. International Journal of Offender Therapy and Comparative Criminology, 63(4), 558–573. https://doi.org/10.1177/0306624X18803818

Cheung, M. W.-L. (2014). Modeling dependent effect sizes with three-level meta-analyses: A structural equation modeling approach. Psychological Methods, 19(2), 211–229. https://doi.org/10.1037/a0032968

Chow, J. C., Wallace, E. S., Senter, R., Kumm, S., & Mason, C. Q. (2022). A Systematic Review and Meta-Analysis of the Language Skills of Youth Offenders. Journal of Speech, Language, and Hearing Research, 65(3), 1166–1182. https://doi.org/10.1044/2021_JSLHR-20-00308

Choy, O., & Raine, A. (2024). The neurobiology of antisocial personality disorder. Neuropharmacology, 261, 110150. https://doi.org/10.1016/j.neuropharm.2024.110150

Chun, M. M., & Turk-Browne, N. B. (2007). Interactions between attention and memory. Current Opinion in Neurobiology, 17(2), 177–184. https://doi.org/10.1016/j.conb.2007.03.005

*Cohen, L. J., Nesci, C., Steinfeld, M., Haeri, S., & Galynker, I. (2010). Investigating the Relationship between Sexual and Chemical Addictions by Comparing Executive Function in Subjects With Pedophilia or Opiate Addiction and Healthy Controls. Journal of Psychiatric Practice, 16(6), 405–412. https://doi.org/10.1097/01.pra.0000390759.04581.7c

*Cohen, R. A., Brumm, V., Zawacki, T. M., Paul, R., Sweet, L., & Rosenbaum, A. (2003). Impulsivity and verbal deficits associated with domestic violence. Journal of the

International Neuropsychological Society, 9(5), 760–770. https://doi.org/10.1017/S1355617703950090

Cohen, J. (2009). Statistical power analysis for the behavioral sciences. 2nd ed. Psychology Press.

Cohen, R. A., Brumm, V., Zawacki, T. M., Paul, R., Sweet, L., & Rosenbaum, A. (2003). Impulsivity and verbal deficits associated with domestic violence. Journal of the International Neuropsychological Society, 9(5), 760–770. https://doi.org/10.1017/S1355617703950090

*Combalbert, N. (2018). Cognitive impairment, self-perceived health and quality of life of older prisoners. Criminal Behaviour and Mental Health, 28(1), 36–49. https://doi.org/10.1002/cbm.2023

*Combalbert, N., Pennequin, V., Ferrand, C., Vandevyvère, R., Armand, M., & Geffray, B. (2016). Mental disorders and cognitive impairment in ageing offenders. The Journal of Forensic Psychiatry & Psychology, 27(6), 853–866. https://doi.org/10.1080/14789949.2016.1244277

Cortés Pascual, A., Moyano Muñoz, N., & Quílez Robres, A. (2019). The Relationship Between Executive Functions and Academic Performance in Primary Education: Review and Meta-Analysis. Frontiers in Psychology, 10, 1582. https://doi.org/10.3389/fpsyg.2019.01582

Crick, N. R., & Dodge, K. A. (1994). A review and reformulation of social information-processing mechanisms in children’s social adjustment. Psychological Bulletin, 115(1), 74–101. https://doi.org/10.1037/0033-2909.115.1.74

Cruz, A. R., de Castro-Rodrigues, A., & Barbosa, F. (2020). Executive dysfunction, violence and aggression. Aggression and Violent Behavior, 51, 101380. https://doi.org/10.1016/j.avb.2020.101380

*Curtis, A., Gooden, J. R., Cox, C. A., Harries, T., Peterson, V., Enticott, P. G., Sanfilippo, P. G., Miller, P. G., Lubman, D. I., & Manning, V. (2021). Neurocognitive functioning among people accessing an addiction neuropsychology clinic with and without a history of offending behaviour. Psychiatry Psychology and Law, 28(6), 854–866. https://doi.org/10.1080/13218719.2021.1873204

*De Brito, S. A., Viding, E., Kumari, V., Blackwood, N., & Hodgins, S. (2013). Cool and Hot Executive Function Impairments in Violent Offenders with Antisocial Personality Disorder with and without Psychopathy. PLoS ONE, 8(6), e65566. https://doi.org/10.1371/journal.pone.0065566

De Brito, S. A., Viding, E., Kumari, V., Blackwood, N., & Hodgins, S. (2013). Cool and Hot Executive Function Impairments in Violent Offenders with Antisocial Personality Disorder with and without Psychopathy. PLoS ONE, 8(6), e65566. https://doi.org/10.1371/journal.pone.0065566

*De Girolamo, G., Iozzino, L., Ferrari, C., Gosek, P., Heitzman, J., Salize, H., Wancata, J., Picchioni, M., & Macis, A. (2023). A multinational case-control study comparing forensic and non-forensic patients with schizophrenia spectrum disorders: The EU-VIORMED project. PSYCHOLOGICAL MEDICINE, 53(5), 1814–1824. https://doi.org/10.1017/S0033291721003433

De Vogel, V., Stam, J., Bouman, Y. H. A., Ter Horst, P., & Lancel, M. (2022). Gender Differences in Substance Abuse History and Offending Behavior: A Multicentre Study in Dutch Forensic Psychiatry. Journal of Forensic Psychology Research and Practice, 22(1), 1–17. https://doi.org/10.1080/24732850.2021.1945833

Diamond, A. (2013). Executive Functions. Annual Review of Psychology, 64(1), 135–168. https://doi.org/10.1146/annurev-psych-113011-143750

*Dolan, M., & Anderson, I. M. (2002). Executive and memory function and its relationship to trait impulsivity and aggression in personality disordered offenders. Journal of Forensic Psychiatry, 13(3), 503–526. https://doi.org/10.1080/0958518021000019452

*Dolan, M., Deakin, W. J. F., Roberts, N., & Anderson, I. (2002). Serotonergic and cognitive impairment in impulsive aggressive personality disordered offenders: Are there implications for treatment? Psychological Medicine, 32(1), 105–117. https://doi.org/10.1017/s0033291701004688

*Dolan, M. (2012). The neuropsychology of prefrontal function in antisocial personality disordered offenders with varying degrees of psychopathy. Psychological Medicine, 42(8), 1715–1725. https://doi.org/10.1017/S0033291711002686

*Domes, G., Hollerbach, P., Vohs, K., Mokros, A., & Habermeyer, E. (2013). Emotional Empathy and Psychopathy in Offenders: An Experimental Study. Journal of Personality Disorders, 27(1), 67–84. https://doi.org/10.1521/pedi.2013.27.1.67

Duval, S., & Tweedie, R. (2000). Trim and Fill: A Simple Funnel-Plot-Based Method of Testing and Adjusting for Publication Bias in Meta-Analysis. Biometrics, 56(2), 455–463. https://doi.org/10.1111/j.0006-341X.2000.00455.x

*Easton, C. J., Sacco, K. A., Neavins, T. M., Wupperman, P., & George, T. P. (2008). Neurocognitive Performance Among Alcohol Dependent Men With and Without Physical Violence Toward Their Partners: A Preliminary Report. The American Journal of Drug and Alcohol Abuse, 34(1), 29–37. https://doi.org/10.1080/00952990701764326

*Elsegood, K. J., & Duff, S. C. (2010). Theory of mind in men who have sexually offended against children: A U.K. Comparison study between child sex offenders and nonoffender controls. Sexual Abuse: Journal of Research and Treatment, 22(1), 112–131. https://doi.org/10.1177/1079063209359926

Engel, G. L. (1977). The Need for a New Medical Model: A Challenge for Biomedicine. *Science*, *196*(4286), 129–136. https://doi.org/10.1126/science.847460

*Engelstad, K. N., Rund, B. R., Torgalsbøen, A.-K., Lau, B., Ueland, T., & Vaskinn, A. (2019). Large social cognitive impairments characterize homicide offenders with schizophrenia. Psychiatry Research, 272, 209–215. https://doi.org/10.1016/j.psychres.2018.12.087

*Enticott, P. G., Ogloff, J. R. P., Bradshaw, J. L., & Daffern, M. (2007). Contrary to popular belief, a lack of behavioural inhibitory control may not be associated with aggression. Criminal Behaviour and Mental Health, 17(3), 179–183. https://doi.org/10.1002/cbm.645

*Enticott, P. G., Ogloff, J. R. P., Bradshaw, J. L., & Fitzgerald, P. B. (2008). Cognitive inhibitory control and self-reported impulsivity among violent offenders with schizophrenia. Journal of Clinical and Experimental Neuropsychology, 30(2), 1–6. https://doi.org/10.1080/13803390701290055

Farrer, T. J., & Hedges, D. W. (2011). Prevalence of traumatic brain injury in incarcerated groups compared to the general population: A meta-analysis. Progress in Neuro-Psychopharmacology and Biological Psychiatry, 35(2), 390–394. https://doi.org/10.1016/j.pnpbp.2011.01.007

Fayyad, J., Sampson, N. A., Hwang, I., Adamowski, T., Aguilar-Gaxiola, S., Al-Hamzawi, A., Andrade, L. H. S. G., Borges, G., de Girolamo, G., Florescu, S., Gureje, O., Haro, J. M., Hu, C., Karam, E. G., Lee, S., Navarro-Mateu, F., O'Neill, S., Pennell, B.-E., Piazza, M., … Kessler, R. C. (2017). The descriptive epidemiology of DSM-IV Adult ADHD in the World Health Organization World Mental Health Surveys. ADHD Attention Deficit and Hyperactivity Disorders, 9(1), 47–65. https://doi.org/10.1007/s12402-016-0208-3

Fazel, S., & Danesh, J. (2002). Serious mental disorder in 23 000 prisoners: A systematic review of 62 surveys. The Lancet, 359(9306), 545–550. https://doi.org/10.1016/S0140-6736(02)07740-1

Fazel, S., Yoon, I. A., & Hayes, A. J. (2017). Substance use disorders in prisoners: An updated systematic review and meta-regression analysis in recently incarcerated men and women. Addiction, 112(10), 1725–1739. https://doi.org/10.1111/add.13877

Fernández-Castilla, B., Declercq, L., Jamshidi, L., Beretvas, S. N., Onghena, P., & Van den Noortgate, W. (2020). Visual representations of meta-analyses of multiple outcomes: Extensions to forest plots, funnel plots, and caterpillar plots. Methodology, 16(4), 299–315. https://doi.org/10.5964/meth.4013

Fernández-Castilla, B., Declercq, L., Jamshidi, L., Beretvas, S. N., Onghena, P., & Van den Noortgate, W. (2021). Detecting Selection Bias in Meta-Analyses with Multiple Outcomes: A Simulation Study. The Journal of Experimental Education, 89(1), 125–144. https://doi.org/10.1080/00220973.2019.1582470

Fernández-Castilla, B., Jamshidi, L., Declercq, L., Beretvas, S. N., Onghena, P., & Van den Noortgate, W. (2020). The application of meta-analytic (multi-level) models with multiple random effects: A systematic review. Behavior Research Methods, 52(5), 2031–2052. https://doi.org/10.3758/s13428-020-01373-9

*Franke, I., Seipel, S., Vasic, N., Streb, J., Nigel, S., Otte, S., & Dudeck, M. (2019). Neuropsychological profile of pedophilic child sexual offenders compared with an IQ-matched non-offender sample – Results of a pilot study. International Journal of Law and Psychiatry, 64, 137–141. https://doi.org/10.1016/j.ijlp.2019.03.002

Frost, R. B., Farrer, T. J., Primosch, M., & Hedges, D. W. (2013). Prevalence of Traumatic Brain Injury in the General Adult Population: A Meta-Analysis. Neuroepidemiology, 40(3), 154–159. https://doi.org/10.1159/000343275

Fuermaier, A. B. M., Fricke, J. A., de Vries, S. M., Tucha, L., & Tucha, O. (2019). Neuropsychological assessment of adults with ADHD: A Delphi consensus study. Applied Neuropsychology: Adult, 26(4), 340–354. https://doi.org/10.1080/23279095.2018.1429441

Ginsberg, Y., Hirvikoski, T., & Lindefors, N. (2010). Attention Deficit Hyperactivity Disorder (ADHD) among longer-term prison inmates is a prevalent, persistent and disabling disorder. BMC Psychiatry, 10(1), 112. https://doi.org/10.1186/1471-244X-10-112

*Gobbi, E., Cotelli, M., Manenti, R., Ferrari, C., Macis, A., Bianconi, G., Candini, V., Clerici, M., Ferla, M. T., Iozzino, L., Vita, A., & de Girolamo, G. (2020). Neuropsychological features in patients with severe mental disorders and risk of violence: A prospective multicenter study in Italy. Psychiatry Research, 289, 113027. https://doi.org/10.1016/j.psychres.2020.113027

Gonzalez Aguilar, M. J. (2021). Cognition, Emotion and Behaviour: Addressing the Relationship Between Executive Functions and Decision-Making. International Journal on Neuropsychology and Behavioural Sciences (IJNBS), 2(4). https://doi.org/10.51626/ijnbs.2021.02.00020

*Grant, J. E., Chamberlain, S. R., Schreiber, L. R. N., & Odlaug, B. L. (2012). Neurocognitive deficits associated with shoplifting in young adults. Comprehensive Psychiatry, 53(8), 1049–1055. https://doi.org/10.1016/j.comppsych.2012.04.012

*Greenfield, R., & Valliant, P. M. (2007). Moral Reasoning, Executive Function, and Personality in Violent and Nonviolent Adult Offenders. Psychological Reports, 101, 323–333. https://doi.org/10.2466/pr0.101.1.323-333

Guo, N., Fuermaier, A. B. M., Koerts, J., Tucha, O., Scherbaum, N., & Müller, B. W. (2023). Networks of Neuropsychological Functions in the Clinical Evaluation of Adult ADHD. Assessment, 30(6), 1719–1736. https://doi.org/10.1177/10731911221118673

Haarsma, G., Davenport, S., White, D. C., Ormachea, P. A., Sheena, E., & Eagleman, D. M. (2020). Assessing Risk Among Correctional Community Probation Populations: Predicting Reoffense With Mobile Neurocognitive Assessment Software. Frontiers in Psychology, 10, 2926. https://doi.org/10.3389/fpsyg.2019.02926

*Habermeyer, B., Esposito, F., Händel, N., Lemoine, P., Kuhl, H. C., Klarhöfer, M., Mager, R., Mokros, A., Dittmann, V., Seifritz, E., & Graf, M. (2013). Response Inhibition in Pedophilia: An fMRI Pilot Study. Neuropsychobiology, 68(4), 228–237. https://doi.org/10.1159/000355295

*Hanlon, R. E., Coda, J. J., Cobia, D., & Rubin, L. H. (2012). Psychotic domestic murder: Neuropsychological differences between homicidal and nonhomicidal schizophrenic men. Journal of Family Violence, 27(2), 105–113. https://doi.org/DOI:10.1007/s10896-011-9410-4

Harris, J. C. (2006). Intellectual disability: Understanding its development, causes, classification, evaluation, and treatment. Oxford University Press.

Hellenbach, M., Karatzias, T., & Brown, M. (2017). Intellectual Disabilities Among Prisoners: Prevalence and Mental and Physical Health Comorbidities. Journal of Applied Research in Intellectual Disabilities, 30(2), 230–241. https://doi.org/10.1111/jar.12234

*Herrero, O., Escorial, S., & Colom, R. (2019). Rapists and child abusers share low levels in executive updating, but do not in fluid reasoning. The European Journal of Psychology Applied to Legal Context, 11(1), 1–7. https://dx.doi.org/10.5093/ejpalc2018a10

Higgins, J. P. T. (2003). Measuring inconsistency in meta-analyses. BMJ, 327(7414), 557–560. https://doi.org/10.1136/bmj.327.7414.557

Higgins, J. P. T., & Thompson, S. G. (2002). Quantifying heterogeneity in a meta-analysis. Statistics in Medicine, 21(11), 1539–1558. https://doi.org/10.1002/sim.1186

*Högman, L., Gavalova, G., Laukka, P., Kristiansson, M., Källman, M. V., Fischer, H., & Johansson, A. G. M. (2023). Cognition, prior aggression, and psychopathic traits in relation to impaired multimodal emotion recognition in psychotic spectrum disorders. Frontiers in Psychiatry, 14, 1111896. https://doi.org/10.3389/fpsyt.2023.1111896

*Hoppenbrouwers, S. S., De Jesus, D. R., Stirpe, T., Fitzgerald, P. B., Voineskos, A. N., Schutter, D. J. L. G., & Daskalakis, Z. J. (2013). Inhibitory deficits in the dorsolateral prefrontal cortex in psychopathic offenders. Cortex: A Journal Devoted to the Study of the Nervous System and Behavior, 49(5), 1377–1385. https://doi.org/10.1016/j.cortex.2012.06.003

Hox, J. J., Moerbeek, M., & Schoot, R. van de. (2018). Multilevel analysis: Techniques and applications (Third edition). Routledge, Taylor & Francis Group.

Humenik, A. M., Grounds, Z. K., Mayer, H. M., & Dolan, S. L. (2020). A systematic review of executive cognitive function in intimate partner violent offenders. Aggression and Violent Behavior, 54, 101407. https://doi.org/10.1016/j.avb.2020.101407

Hunter, J. E., & Schmidt, F. L. (1990). Methods of meta-analysis: Correcting error and bias in research findings. Sage.

IBM Corp. (2023). IBM SPSS Statistics for Windows, Version 29.0.0 [Computer software]. IBM Corp.

*Iozzino, L., Harvey, P. D., Canessa, N., Gosek, P., Heitzman, J., Macis, A., Picchioni, M., Salize, H. J., Wancata, J., Koch, M., Ferrari, C., & de Girolamo, G. (2021). Neurocognition and social cognition in patients with schizophrenia spectrum disorders with and without a history of violence: Results of a multinational European study. Translational Psychiatry, 8(11), 620. https://doi.org/10.1038/s41398-021-01749-1

Janes, S., McIntosh, L. G., O'Rourke, S., & Schwannauer, M. (2024). Examining the cognitive contributors to violence risk in forensic samples: A systematic review and meta-analysis. Aggression and Violent Behavior, 74, 101887. https://doi.org/10.1016/j.avb.2023.101887

Jansen, J. M., & Franse, M. E. (2024). Executive functioning in antisocial behavior: A multi-level systematic meta-analysis. Clinical Psychology Review, 109, 102408. https://doi.org/10.1016/j.cpr.2024.102408

*Jones, C. G. A., Fearnley, H., Panagiotopoulos, B., & Kemp, R. I. (2015). Delay discounting, self-control, and substance use among adult drug court participants. Behavioural Pharmacology, 26(5), 447–459. https://doi.org/10.1097/FBP.0000000000000149

*Kärgel, C., Massau, C., Weiß, S., Walter, M., Borchardt, V., Krueger, T. H. C., Tenbergen, G., Kneer, J., Wittfoth, M., Pohl, A., er, Gerwinn, H., Ponseti, J., Amelung, T., Beier, K. M., Mohnke, S., Walter, H., & Schiffer, B. (2017). Evidence for superior neurobiological and behavioral inhibitory control abilities in non-offending as compared to offending pedophiles. Human Brain Mapping, 38(2), 1092–1104. https://doi.org/10.1002/hbm.23443

Karoğlu, N., Ferguson, H. J., & Ó Ciardha, C. (2022). Theory of Mind in Offending: A Systematic Review. Trauma, Violence, & Abuse, 23(5), 1610–1628. https://doi.org/10.1177/15248380211013143

*Kashiwagi, H., Kuroki, N., Ikezawa, S., Matsushita, M., Ishikawa, M., Nakagome, K., Hirabayashi, N., & Ikeda, M. (2015). Neurocognitive features in male patients with schizophrenia exhibiting serious violence: A case control study. Annals of General Psychiatry, 14(1), 46. https://doi.org/10.1186/s12991-015-0086-7

Kazeem, O. T. (2020). Adverse Childhood Experiences, Socio-Economic Status, and Criminal Behaviour: A Cross-Sectional Correctional Survey. Adversity and Resilience Science, 1(4), 319–327. https://doi.org/10.1007/s42844-020-00018-2

Klin, A. (2000). Attributing Social Meaning to Ambiguous Visual Stimuli in Higher-functioning Autism and Asperger Syndrome: The Social Attribution Task. Journal of Child Psychology and Psychiatry, 41(7), 831–846. https://doi.org/10.1111/1469-7610.00671

Knapp, G., & Hartung, J. (2003). Improved tests for a random effects meta-regression with a single covariate. Statistics in Medicine, 22(17), 2693–2710. https://doi.org/10.1002/sim.1482

*Krakowski, M. I., Foxe, J., de Sanctis, P., Nolan, K., Hoptman, M. J., Shope, C., Kamiel, S., & Czobor, P. (2015). Aberrant response inhibition and task switching in psychopathic individuals. Psychiatry Research, 229(3), 1017–1023. https://doi.org/10.1016/j.psychres.2015.06.018

*Lahat, A., Gummerum, M., Mackay, L., & Hanoch, Y. (2015). Cognitive processing of moral and social judgements: A comparison of offenders, students, and control participants. The Quarterly Journal of Experimental Psychology, 68(2), 350–362. https://doi.org/10.1080/17470218.2014.944918

Lakens, D. (2013). Calculating and reporting effect sizes to facilitate cumulative science: A practical primer for t-tests and ANOVAs. Frontiers in Psychology, 4. https://doi.org/10.3389/fpsyg.2013.00863

Lenzenweger, M. F., Lane, M. C., Loranger, A. W., & Kessler, R. C. (2007). DSM-IV Personality Disorders in the National Comorbidity Survey Replication. Biological Psychiatry, 62(6), 553–564. https://doi.org/10.1016/j.biopsych.2006.09.019

*Lishak, V., Scott, K. L., Dyson, A., & Milovanov, A. (2019). General Criminality as a Marker of Heterogeneity in Domestically Violent Men: Differences in Trauma History, Psychopathology, Neurocognitive Functioning, and Psychophysiology. Journal of Interpersonal Violence, 088626051985806. https://doi.org/10.1177/0886260519858065

Lukito, S., Norman, L., Carlisi, C., Radua, J., Hart, H., Simonoff, E., & Rubia, K. (2020). Comparative meta-analyses of brain structural and functional abnormalities during cognitive control in attention-deficit/hyperactivity disorder and autism spectrum disorder. Psychological Medicine, 50(6), 894–919. https://doi.org/10.1017/S0033291720000574

*Majorek, K., Wolfkuhler, W., Kuper, C., Saimeh, N., Juckel, G., & Brune, M. (2009). Theory of mind" and executive functioning in forensic patients with schizophrenia. Journal of Forensic Sciences, 54, 469–473. https://doi.org/10.1111/j.1556-4029.2008.00966.x

*Marceau, R., Meghani, R., & Reddon, J. R. (2008). Neuropsychological assessment of adult offenders. Special Issue: Offender Assessment., 47(1), 41–73. https://psycnet.apa.org/doi/10.1080/10509670801940409

*Mariani, F., Calandri, I., & Dansilio, S. (2024). Criminal behaviors: A theory of mind problem? Applied Neuropsychology: Adult, 1–10. APA PsycInfo. https://doi.org/10.1080/23279095.2024.2326935

*Mariano, M., Pino, M. C., Peretti, S., Valenti, M., & Mazza, M. (2017). Understanding criminal behavior: Empathic impairment in criminal offenders. Social Neuroscience, 12(4), 379–385. https://doi.org/10.1080/17470919.2016.1179670

Marsh, A. A., & Blair, R. J. R. (2008). Deficits in facial affect recognition among antisocial populations: A meta-analysis. Neuroscience and Biobehavioral Reviews, 32(3), 454–465. https://doi.org/10.1016/j.neubiorev.2007.08.003

*Massau, C., Tenbergen, G., Kargel, C., Weis, S., Gerwinn, H., Pohl, A., Amelung, T., Mohnke, S., Kneer, J., Wittfoth, M., Ristow, I., Schiltz, K., Beier, K. M., Ponseti, J., Walter, M., Kruger, T. H. C., Walter, H., & Schiffer, B. (2017). Executive functioning in pedophilia and child sexual offending. Journal of the International Neuropsychological Society, 23(6), 460–470. https://doi.org/10.1017/s1355617717000315

Matheson, F. I., Dastoori, P., Whittingham, L., Calzavara, A., Keown, L. A., Durbin, A., Kouyoumdjian, F. G., Lin, E., Volpe, T., & Lunsky, Y. (2022). Intellectual/developmental disabilities among people incarcerated in federal correctional facilities in Ontario, Canada: Examining prevalence, health and correctional characteristics. Journal of Applied Research in Intellectual Disabilities, 35(3), 900–909. https://doi.org/10.1111/jar.12995

*Mayer, S. V., Jusyte, A., Klimecki-Lenz, O. M., & Schoenenberg, M. (2018). Empathy and altruistic behavior in antisocial violent offenders with psychopathic traits. Psychiatry Research, 269, 625–632. https://doi.org/10.1016/j.psychres.2018.08.035

*Meier, N. M., Perrig, W., & Koenig, T. (2012). Neurophysiological correlates of delinquent behaviour in adult subjects with ADHD. International Journal of Psychophysiology, 84(1), 1–16. https://doi.org/10.1016/j.ijpsycho.2011.12.011

*Molleman, P. W., Driessen, J. M. A., Schilder, C., Bulten, B. H., & Brazil, I. A. (2021). Behavioral Inhibition and Activation System Factors in Offenders and Non-Offenders. International Journal of Forensic Mental Health, 1–13. https://doi.org/10.1080/14999013.2021.1947422

*Molleman, P. W., Molleman, L., Schilder, C., Bulten, B. H., & Brazil, I. A. (2022). Can measures of cognitive flexibility and inhibition distinguish forensic psychiatric inpatients from prisoners? The Journal of Forensic Psychiatry & Psychology, 1–18. https://doi.org/10.1080/14789949.2022.2070523

*Molleman, P. W., Smeekens, M. V., Sappelli, F., Kessels, R. P. C., Brazil, I. A., & Bulten, B. H. (2024). Charting cognitive characteristics in forensic psychiatric outpatients. Psychology & Neuroscience. https://doi.org/10.1037/pne0000337

Molleman, P. W., Smeekens, M. V., Sappelli, F., Kessels, R. P. C., Brazil, I. A., & Bulten, B. H. (2024). Charting cognitive characteristics in forensic psychiatric outpatients. Psychology & Neuroscience. https://doi.org/10.1037/pne0000337

Monahan, J., & Skeem, J. L. (2014). Risk Redux: The Resurgence of Risk Assessment in Criminal Sanctioning. Federal Sentencing Reporter, 26(3), 158–166. https://doi.org/10.1525/fsr.2014.26.3.158

Morgan, A. B., & Lilienfeld, S. O. (2000). A meta-analytic review of the relation between antisocial behavior and neuropsychological measures of executive function. Clinical Psychology Review, 20(1), 113–136. https://doi.org/10.1016/S0272-7358(98)00096-8

Muñoz García-Largo, L., Martí-Agustí, G., Martin-Fumadó, C., & Gómez-Durán, E. L. (2020). Intellectual disability rates among male prison inmates. International Journal of Law and Psychiatry, 70, 101566. https://doi.org/10.1016/j.ijlp.2020.101566

*Munro, G. E. S., Dywan, J., Harris, G. T., McKee, S., Unsal, A., & Segalowitz, S. J. (2007b). Response inhibition in psychopathy: The frontal N2 and P3. Neuroscience Letters, 418(2), 149–153. https://doi.org/10.1016/j.neulet.2007.03.017

*Munro, G. E. S., Dywan, J., Harris, G. T., McKee, S., Unsal, A., & Segalowitz, S. J. (2007a). ERN varies with degree of psychopathy in an emotion discrimination task. Biological Psychology, 76(1), 31–42. https://doi.org/10.1016/j.biopsycho.2007.05.004

Nauta-Jansen, L. M. C. (2022). The neurobiology of antisocial behavior in adolescence; current knowledge and relevance for youth forensic clinical practice. Current Opinion in Psychology, 47, 101356. https://doi.org/10.1016/j.copsyc.2022.101356

*Nentjes, L., Bernstein, D. P., Arntz, A., Slaats, M. E., & Hannemann, T. (2015). Theory of mind, social desirability, and unlikely symptom reporting in offenders with and without psychopathy. Journal of Nervous and Mental Disease, 203(8), 596–603. https://doi.org/10.1097/nmd.0000000000000335

*Neves, D., & Pinho, M. S. (2016). Specificity and emotional characteristics of the autobiographical memories of male and female criminal offenders. Criminal Justice and Behavior, 43(5), 670–690. https://doi.org/10.1177/0093854815613820

*Neves, D., & Pinho, M. S. (2018). Self-regulation and the specificity of autobiographical memory in offenders. International Journal of Law and Psychiatry, 57, 91–99. https://doi.org/10.1016/j.ijlp.2018.01.005

*Newbury-Helps, J., Feigenbaum, J., & Fonagy, P. (2017). Offenders with antisocial personality disorder display more impairments in mentalizing. Journal of Personality Disorders, 31(2), 232–255. https://doi.org/10.1521/pedi_2016_30_246

Noll, K. R., Bradshaw, M. E., Rexer, J., & Wefel, J. S. (2018). Neuropsychological Practice in the Oncology Setting. Archives of Clinical Neuropsychology, 33(3), 344–353. https://doi.org/10.1093/arclin/acx131

Norman, E. M., Polaschek, D. L. L., & Starkey, N. J. (2023). Executive function in individuals who are compliant and non-compliant with the conditions of a community-based sentence. Psychiatry, Psychology and Law, 30(2), 161–176. https://doi.org/10.1080/13218719.2021.2003268

Nuechterlein, K. H., Green, M. F., & Kern, R. S. (2022). The MATRICS Consensus Cognitive Battery: An Update. In D. M. Barch & J. W. Young (Eds.), Cognitive Functioning in Schizophrenia: Leveraging the RDoC Framework (Vol. 63, pp. 1–18). Springer International Publishing. https://doi.org/10.1007/7854_2022_395

Ogilvie, J. M., Stewart, A. L., Chan, R. C. K., & Shum, D. H. K. (2011). Neuropsychological measures of executive function and antisocial behavior: A meta-analysis. Criminology, 49(4), 1063–1107. https://doi.org/10.1111/j.1745-9125.2011.00252.x

Ogonah, M. G. T., Seyedsalehi, A., Whiting, D., & Fazel, S. (2023). Violence risk assessment instruments in forensic psychiatric populations: A systematic review and meta-analysis. The Lancet Psychiatry, 10(10), 780–789. https://doi.org/10.1016/S2215-0366(23)00256-0

Olver, M. E., Stockdale, K. C., & Wormith, J. S. (2011). A meta-analysis of predictors of offender treatment attrition and its relationship to recidivism. Journal of Consulting and Clinical Psychology, 79(1), 6–21. https://doi.org/10.1037/a0022200

*Ostrosky, F., Diaz, Romero, & Borja. (2012). Neuropsychological profiles of members of organized crime and drug-traffic organizations. Research and Reports in Forensic Medical Science, 19. https://doi.org/10.2147/RRFMS.S32352

Ouzzani, M., Hammady, H., Fedorowicz, Z., & Elmagarmid, A. (2016). Rayyan—A web and mobile app for systematic reviews. Systematic Reviews, 5(210). https://doi.org/10.1186/s13643-016-0384-4

Page, M. J., McKenzie, J. E., Bossuyt, P. M., Boutron, I., Hoffmann, T. C., Mulrow, C. D., Shamseer, L., Tetzlaff, J. M., Akl, E. A., Brennan, S. E., Chou, R., Glanville, J., Grimshaw, J. M., Hróbjartsson, A., Lalu, M. M., Li, T., Loder, E. W., Mayo-Wilson, E., McDonald, S., … Moher, D. (2021). The PRISMA 2020 statement: An updated guideline for reporting systematic reviews. BMJ, n71. https://doi.org/10.1136/bmj.n71

*Palix, J., Gillespie, S. M., Abbiati, M., & Abu-Akel, A. (2022). Dynamics of parasympathetic activity in violent incarcerated offenders before, during, and in recovery from an emotional inhibition task. Scientific Reports, 12(1), 7126. https://doi.org/10.1038/s41598-022-10872-y

*Pasion, R., Cruz, A. R., & Barbosa, F. (2018). Dissociable effects of psychopathic traits on executive functioning: Insights from the triarchic model. Frontiers in Psychology, 9, 1–10. https://doi.org/10.3389/fpsyg.2018.01713

*Pera-Guardiola, V., Batalla, I., Bosque, J., Kosson, D., Pifarre, J., Hernández-Ribas, R., Goldberg, X., Contreras-Rodriguez, O., Menchon, J. M., Soriano-Mas, C., & Cardoner, N. (2016). Modulatory effects of psychopathy on Wisconsin Card Sorting Test performance in male offenders with Antisocial Personality Disorder. Psychiatry Research, 235, 43–48. https://doi.org/10.1016/j.psychres.2015.12.003

*Prehn, K., Schlagenhauf, F., Schulze, L., Berger, C., Vohs, K., Fleischer, M., Hauenstein, K., Keiper, P., Domes, G., & Herpertz, S. C. (2013). Neural correlates of risk taking in violent criminal offenders characterized by emotional hypo- and hyper-reactivity. Social Neuroscience, 8(2), 136–147. https://doi.org/10.1016/j.psychres.2015.12.003

*Proctor, T., & Beail, N. (2007). Empathy and theory of mind in offenders with intellectual disability. Special Issue: Trends and Challenges in Forensic Research on Offenders with Intellectual Disability, 32(2), 82–93. https://doi.org/10.1080/13668250701373331

R Core Team. (2021). R: A language and environment for statistical computing [Computer software]. R Foundation for Statistical Computing. https://www.R-project.org/.

Raine, A., Moffitt, T. E., Caspi, A., Loeber, R., Stouthamer-Loeber, M., & Lynam, D. (2005). Neurocognitive Impairments in Boys on the Life-Course Persistent Antisocial Path. Journal of Abnormal Psychology, 114(1), 38–49. https://doi.org/10.1037/0021-843X.114.1.38

Raitano Lee, N., Maiman, M., & Godfrey, M. (2016). What can Neuropsychology Teach Us About Intellectual Disability? In International Review of Research in Developmental Disabilities (Vol. 51, pp. 1–40). Elsevier. https://doi.org/10.1016/bs.irrdd.2016.07.002

*Robinson, L., Spencer, M. D., Thomson, L. D. G., Sprengelmeyer, R., Owens, D. G. C., Stanfield, A. C., Hall, J., Baig, B. J., MacIntyre, D. J., McKechanie, A., & Johnstone, E. C. (2012). Facial emotion recognition in Scottish prisoners. International Journal of Law and Psychiatry, 35(1), 57–61. https://doi.org/10.1016/j.ijlp.2011.11.009

Roest, J. J., Welmers-Van de Poll, M. J., Peer Van der Helm, G. H., Stams, G. J. J. M., & Hoeve, M. (2022). A Three-level Meta-analysis on the Alliance-Outcome Association in Child and Adolescent Psychotherapy. Research on Child and Adolescent Psychopathology. https://doi.org/10.1007/s10802-022-00986-2

*Romero-Martínez, Á., Lila, M., & Moya-Albiol, L. (2019a). Long-Term Drug Misuse Increases the Risk of Cognitive Dysfunctions in Intimate Partner Violence Perpetrators: Key Intervention Targets for Reducing Dropout and Reoffending. International Journal of Environmental Research and Public Health, 16(20), 3792. https://doi.org/10.3390/ijerph16203792

*Romero-Martinez, A., Lila, M., & Moya-Albiol, L. (2019b). The importance of impulsivity and attention switching deficits in perpetrators convicted for intimate partner violence. Aggressive Behavior, 45(2), 129–138. https://doi.org/10.1002/ab.21802

*Romero-Martinez, A., Lila, M., & Moya-Albiol, L. (2021). Alexithymic traits are closely related to impulsivity and cognitive and empathic dysfunctions in intimate partner violence perpetrators: New targets for intervention. Applied Neuropsychology: Adult, 28(1), 71–79. https://doi.org/10.1080/23279095.2019.1594233

*Romero-Martinez, A., Lila, M., Sarinana-Gonzalez, P., Gonzalez-Bono, E., & Moya-Albiol, L. (2013). High testosterone levels and sensitivity to acute stress in perpetrators of domestic violence with low cognitive flexibility and impairments in their emotional decoding process: A preliminary study. Aggressive Behavior, 39(5), 355–369. https://doi.org/10.1002/ab.21490

*Romero-Martínez, Á., Lila, M., Sarrate-Costa, C., Comes-Fayos, J., & Moya-Albiol, L. (2022). Neuropsychological Performance, Substance Misuse, and Recidivism in Intimate Partner Violence Perpetrators. Psychosocial Intervention, 26(32), 69–77. https://doi.org/10.5093/pi2022a7

*Romero-Martínez, Á., Lila, M., Sarrate-Costa, C., Comes-Fayos, J., & Moya-Albiol, L. (2023a). The Interaction between Attention Deficit Hyperactivity Disorder and Neuropsychological Deficits for Explaining Dropout and Recidivism of Intimate Partner Violence Perpetrators. The European Journal of Psychology Applied to Legal Context, 15(1), 33–42. https://doi.org/10.5093/ejpalc2023a4

*Romero-Martinez, A., Lila, M., Vitoria-Estruch, S., & Moya-Albiol, L. (2021). Can Attention and Working Memory Impairments of Intimate Partner Perpetrators Explain Their Risky Decision Making? Journal of Interpersonal Violence, 36(11–12), NP6492–NP6507. https://doi.org/10.1177/0886260518814263

Romero-Martínez, Á., Lila, M., Sarrate-Costa, C., Comes-Fayos, J., & Moya-Albiol, L. (2023). The Interaction between Attention Deficit Hyperactivity Disorder and Neuropsychological Deficits for Explaining Dropout and Recidivism of Intimate

Partner Violence Perpetrators. The European Journal of Psychology Applied to Legal Context, 15(1), 33–42. https://doi.org/10.5093/ejpalc2023a4

*Rosburg, T., Deuring, G., Boillat, C., Lemoine, P., Falkenstein, M., Graf, M., & Mager, R. (2018). Inhibition and attentional control in pedophilic child sexual offenders—An event-related potential study. Clinical Neurophysiology : Official Journal of the International Federation of Clinical Neurophysiology, 129(9), 1990–1998. https://doi.org/10.1016/j.clinph.2018.06.029

Ross, E. H., & Hoaken, P. N. S. (2011). Executive Cognitive Functioning Abilities of Male First Time and Return Canadian Federal Inmates. Canadian Journal of Criminology and Criminal Justice, 53(4), 377–403. https://doi.org/10.3138/cjccj.53.4.377

*Santos Barbosa, M. F., & Coelho Monteiro, L. M. (2008). Recurrent criminal behavior and executive dysfunction. The Spanish Journal of Psychology, 11(1), 259–265. https://doi.org/10.1017/s1138741600004297

Savage, J., Ferguson, C. J., & Flores, L. (2017). The effect of academic achievement on aggression and violent behavior: A meta-analysis. Aggression and Violent Behavior, 37, 91–101. https://doi.org/10.1016/j.avb.2017.08.002

*Schiffer, B., & Vonlaufen, C. (2011). Executive dysfunctions in pedophilic and nonpedophilic child molesters. Journal of Sexual Medicine, 8(7), 1975–1984. https://doi.org/10.1111/j.1743-6109.2010.02140.x

*Schuler, M., Mohnke, S., Amelung, T., Beier, K. M., Walter, M., Ponseti, J., Schiffer, B., Kruger, T. H. C., & Walter, H. (2022). Neural processing associated with cognitive empathy in pedophilia and child sexual offending. Social Cognitive and Affective Neuroscience, 17(8), 712–722. https://doi.org/10.1093/scan/nsab133

Seguin, J. R., Sylvers, P., & Lilienfeld, S. O. (2007). The Neuropsychology of Violence. In D. J. Flannery, A. T. Vazsonyi, I. D. Waldman, & D. Irwin, The Cambridge handbook of violent behavior and aggression (p. 28).

*Seidel, E.-M., Pfabigan, D. M., Keckeis, K., Wucherer, A. M., Jahn, T., Lamm, C., & Derntl, B. (2013). Empathic competencies in violent offenders. Psychiatry Research, 210(3), 1168–1175. https://doi.org/10.1016/j.psychres.2013.08.027

*Seruca, T., & Silva, C. F. (2016). Executive Functioning in Criminal Behavior: Differentiating between Types of Crime and Exploring the Relation between Shifting, Inhibition, and Anger. International Journal of Forensic Mental Health, 15(3), 235–246. https://doi.org/10.1080/14999013.2016.1158755

*Shamay-Tsoory, S. G., Harari, H., Aharon-Peretz, J., & Levkovitz, Y. (2010). The role of the orbitofrontal cortex in affective theory of mind deficits in criminal offenders with psychopathic tendencies. Cortex: A Journal Devoted to the Study of the Nervous System and Behavior, 46(5), 668–677. https://doi.org/10.1016/j.cortex.2009.04.008

Shumlich, E. J., Reid, G. J., Hancock, M., & Hoaken, P. (2019). Executive Dysfunction in Criminal Populations: Comparing Forensic Psychiatric Patients and Correctional Offenders. International Journal of Forensic Mental Health, 18(3), 243–259. https://doi.org/10.1080/14999013.2018.1495279

*Slotboom, J., Hoppenbrouwers, S. S., Bouman, Y. H. A., in 't Hout, W., Sergiou, C., van der Stigchel, S., & Theeuwes, J. (2017). Visual attention in violent offenders: Susceptibility to distraction. Psychiatry Research, 251, 281–286. https://doi.org/10.1016/j.psychres.2017.02.031

Smeijers, D. (2023). Hostility Bias: A Key Characteristic of Aggressive Behavior. In C. R. Martin, V. R. Preedy, & V. B. Patel (Eds.), Handbook of Anger, Aggression, and

Violence (pp. 559–578). Springer International Publishing. https://doi.org/10.1007/978-3-031-31547-3_34

*Spenser, K. A., Betts, L. R., & Das Gupta, M. (2015). Deficits in Theory of Mind, empathic understanding and moral reasoning: A comparison between young offenders and non-offenders. Psychology Crime & Law, 21(7), 632–647. https://psycnet.apa.org/doi/10.1080/1068316X.2015.1028542

*Spenser, K. A. (2017). Socio-Cognitive Determinants of Offending Behaviour. Nottingham Trent University.

*Stanford, M. S., Conklin, S. M., Helfritz, L. E., & Kockler, T. R. (2007). P3 amplitude reduction and executive function deficits in men convicted of spousal/partner abuse. Personality and Individual Differences, 43(2), 365–375. https://psycnet.apa.org/doi/10.1016/j.paid.2006.12.011

Stephan, R. A., Alhassoon, O. M., Allen, K. E., Wollman, S. C., Hall, M., Thomas, W. J., Gamboa, J. M., Kimmel, C., Stern, M., Sari, C., Dalenberg, C. J., Sorg, S. F., & Grant, I. (2017). Meta-analyses of clinical neuropsychological tests of executive dysfunction and impulsivity in alcohol use disorder. The American Journal of Drug and Alcohol Abuse, 43(1), 24–43. https://doi.org/10.1080/00952990.2016.1206113

Steward, K. A., & Kretzmer, T. (2022). Anosognosia in moderate-to-severe traumatic brain injury: A review of prevalence, clinical correlates, and diversity considerations. The Clinical Neuropsychologist, 36(8), 2021–2040. https://doi.org/10.1080/13854046.2021.1967452

*Stratton, J. (2018). Differences in neuropsychological functioning between homicidal and nonviolent schizophrenia samples. Journal of Forensic Sciences, 63(5), 1435–1443. https://doi.org/10.1111/1556-4029.13750

*Suchy, Y., Whittaker, J. W., Strassberg, D. S., & Eastvold, A. (2009a). Neurocognitive differences between pedophilic and nonpedophilic child molesters. Journal of the International Neuropsychological Society, 15(2), 248–257. https://doi.org/10.1017/S1355617709090353

*Suchy, Y., Whittaker, W. J., Strassberg, D. S., & Eastvold, A. (2009b). Facial and prosodic affect recognition among pedophilic and nonpedophilic criminal child molesters. Sexual Abuse: Journal of Research and Treatment, 21(1), 93–110. https://doi.org/10.1177/1079063208326930

Sweet, J. J. (2024). Forensic Neuropsychology Goes from Nowhere to Almost Everywhere: Lessons Learned During Decades of Practice. *Archives of Clinical Neuropsychology*, acae105. https://doi.org/10.1093/arclin/acae105

Sweet, J. J., Boone, K. B., Denney, R. L., Hebben, N., Marcopulos, B. A., Morgan, J. E., Nelson, N. W., & Westerveld, M. (2022). Forensic neuropsychology: History and current status. *The Clinical Neuropsychologist*, 1–16. https://doi.org/10.1080/13854046.2022.2078740

Sweet, J. J., King, J. H., Malina, A. C., Bergman, M. A., & Simmons, A. (2002). Documenting the Prominence of Forensic Neuropsychology at National Meetings and in Relevant Professional Journals from 1990 to 2000. *The Clinical Neuropsychologist*, *16*(4), 481–494. https://doi.org/10.1076/clin.16.4.481.13914Tabachnick, B. G., Fidell, L. S., & Ullman, J. B. (2019). Using multivariate statistics. Pearson.

*Tasios, K., Douzenis, A., Gournellis, R., & Michopoulos, I. (2024). Empathy and Violence in Schizophrenia and Antisocial Personality Disorder. HEALTHCARE, 12(1). https://doi.org/10.3390/healthcare12010089

*Tiberi, L. A., Gillespie, S. M., Saloppé, X., Vicenzutto, A., & Pham, T. H. (2024). Recognition of dynamic facial expressions of emotions in forensic inpatients who have committed sexual offenses: A signal detection analysis. Frontiers in Psychiatry, 15, 1384789. https://doi.org/10.3389/fpsyt.2024.1384789

Trull, T. J., Jahng, S., Tomko, R. L., Wood, P. K., & Sher, K. J. (2010). Revised NESARC Personality Disorder Diagnoses: Gender, Prevalence, and Comorbidity with Substance Dependence Disorders. Journal of Personality Disorders, 24(4), 412–426. https://doi.org/10.1521/pedi.2010.24.4.412

Turk-Browne, N. B., Golomb, J. D., & Chun, M. M. (2013). Complementary attentional components of successful memory encoding. NeuroImage, 66, 553–562. https://doi.org/10.1016/j.neuroimage.2012.10.053

*Turner, D., Brand, M., Welsch, R., Laier, C., Bockshammer, T., & Rettenberger, M. (2018). Response inhibition and impulsive decision-making in sexual offenders against children. Journal of Abnormal Psychology, 127(5), 471–481. https://doi.org/10.1037/abn0000359

*Turner, D., Roszyk, A., Szumski, F., & Rettenberger, M. (2020). Deviant sexual interests but not antisocial behaviors are associated with deficits in executive functioning in individuals convicted of sexual offenses against children. Neuropsychology, 34(8), 906–916. https://doi.org/10.1037/neu0000703

*Valliant, P. M., Freeston, A., Pottier, D., & Kosmyna, R. (2003). Personality and executive functioning as risk factors in recidivists. Psychological Reports, 92(1), 299–306. https://doi.org/10.2466/pr0.2003.92.1.299

Van den Noortgate, W., López-López, J. A., Marín-Martínez, F., & Sánchez-Meca, J. (2013). Three-level meta-analysis of dependent effect sizes. Behavior Research Methods, 45(2), 576–594. https://doi.org/10.3758/s13428-012-0261-6

Van den Noortgate, W., López-López, J. A., Marín-Martínez, F., & Sánchez-Meca, J. (2015). Meta-analysis of multiple outcomes: A multilevel approach. Behavior Research Methods, 47(4), 1274–1294. https://doi.org/10.3758/s13428-014-0527-2

Van Der Sluys, M. E., Zijlmans, J., Popma, A., Van Der Laan, P. H., Scherder, E. J. A., & Marhe, R. (2020). Neurocognitive predictors of treatment completion and daytime activities at follow-up in multiproblem young adults. Cognitive, Affective, & Behavioral Neuroscience, 20(5), 1103–1121. https://doi.org/10.3758/s13415-020-00822-4

Van Goozen, S. H. M., Langley, K., & Hobson, C. W. (2022). Childhood Antisocial Behavior: A Neurodevelopmental Problem. Annual Review of Psychology, 73(1), 353–377. https://doi.org/10.1146/annurev-psych-052621-045243

*Vaskinn, A., Engelstad, K. N., Torgalsboen, A.-K., & Rund, B. R. (2021). Childhood trauma, social cognition and schizophrenia: Specific association between physical neglect and cognitive theory of mind in homicide offenders. Psychiatry Research, 303, 114093. https://doi.org/10.1016/j.psychres.2021.114093

*Vaskinn, A., Rokicki, J., Bell, C., Tesli, N., Bang, N., Hjell, G., Fischer-Vieler, T., Haukvik, U. K., & Friestad, C. (2024). Violent offending in males with or without schizophrenia: A role for social cognition? Schizophrenia Bulletin, 50(3), 663–672. APA PsycInfo. https://doi.org/10.1093/schbul/sbad151

Verdejo-Garcia, A., & Albein-Urios, N. (2021). Impulsivity traits and neurocognitive mechanisms conferring vulnerability to substance use disorders. Neuropharmacology, 183, 108402. https://doi.org/10.1016/j.neuropharm.2020.108402

Viechtbauer, W. (2010). Conducting Meta-Analyses in R with the metafor Package. Journal of Statistical Software, 36(3). https://doi.org/10.18637/jss.v036.i03

Viechtbauer, W. (2022, August 26). Package ‘metafor’. Meta-Analysis Package for R. Version 3.8-1.

Vijayalakshmi, Dr. V., & Patchainayagi, Dr. S. (2020). Role of Memory in Language Learning –A Review. International Journal of Early Childhood Special Education, 12(2), 68–76. https://doi.org/10.9756/INT-JECSE/V12I2.201057

*Vitoria-Estruch, S., Romero-Martinez, A., Lila, M., & Moya-Albiol, L. (2018). Differential cognitive profiles of intimate partner violence perpetrators based on alcohol consumption. Alcohol, 70, 61–71. https://doi.org/10.1016/j.alcohol.2018.01.006

Von Hippel, W. (2007). Aging, Executive Functioning, and Social Control. Current Directions in Psychological Science, 16(5), 240–244. https://doi.org/10.1111/j.1467-8721.2007.00512.x

Wells, G., Shea, B., O’Connell, D., Peterson, J., Welsch, V., Losos, M., & Tugwell, P. (n.d.). The Newcastle-Ottawa Scale (NOS) for assessing the quality of nonrandomised studies in meta-analysis. The Ottawa Hospital Research Institute. Retrieved April 10, 2024, from https://www.ohri.ca/programs/clinical_epidemiology/oxford.asp

Wickham, H., Chang, W., Henry, L., Lin Pedersen, T., Takahashi, K., Wilke, C., Woo, K., Yutani, H., & Dunnington, D. (2023, December 10). Package ‘ggplot2: Create Elegant Data Visualisations Using the Grammar of Graphics. Version 3.4.4. https://cran.r-project.org/web/packages/ggplot2/ggplot2.pdf

Williams, W. H., Chitsabesan, P., Fazel, S., McMillan, T., Hughes, N., Parsonage, M., & Tonks, J. (2018). Traumatic brain injury: A potential cause of violent crime? The Lancet Psychiatry, 5(10), 836–844. https://doi.org/10.1016/S2215-0366(18)30062-2

*Winter, K., Spengler, S., Bermpohl, F., Singer, T., & Kanske, P. (2017). Social cognition in aggressive offenders: Impaired empathy, but intact theory of mind. Scientific Reports, 7(1), 670. https://doi.org/10.1038/s41598-017-00745-0

*Wolfkuhler, W., Majorek, K., Tas, C., Kuper, C., Saimeh, N., Juckel, G., & Brune, M. (2012). Emotion recognition in pictures of facial affect: Is there a difference between forensic and non-forensic patients with schizophrenia? The European Journal of Psychiatry, 26, 73–85. https://doi.org/10.4321/S0213-61632012000200001

*Xian, Z., Liu, H., Gu, Y., Hu, Z., & Li, G. (2024). EEG biomarkers of behavioral inhibition in patients with depression who committed violent offenses: A Go/NoGo ERP study. Cerebral Cortex, 34(2), 1–12. APA PsycInfo. https://doi.org/10.1093/cercor/bhae010

Yang, Y., & Raine, A. (2009). Prefrontal structural and functional brain imaging findings in antisocial, violent, and psychopathic individuals: A meta-analysis. Psychiatry Research: Neuroimaging, 174(2), 81–88. https://doi.org/10.1016/j.pscychresns.2009.03.012

Zijlmans, J., Marhe, R., Bevaart, F., Van Duin, L., Luijks, M.-J. A., Franken, I., Tiemeier, H., & Popma, A. (2021). The predictive value of neurobiological measures for recidivism in delinquent male young adults. Journal of Psychiatry and Neuroscience, 46(2), 10. https://doi.org/10.1503/jpn.200103

Figure 1

PRISMA flow chart of included studies

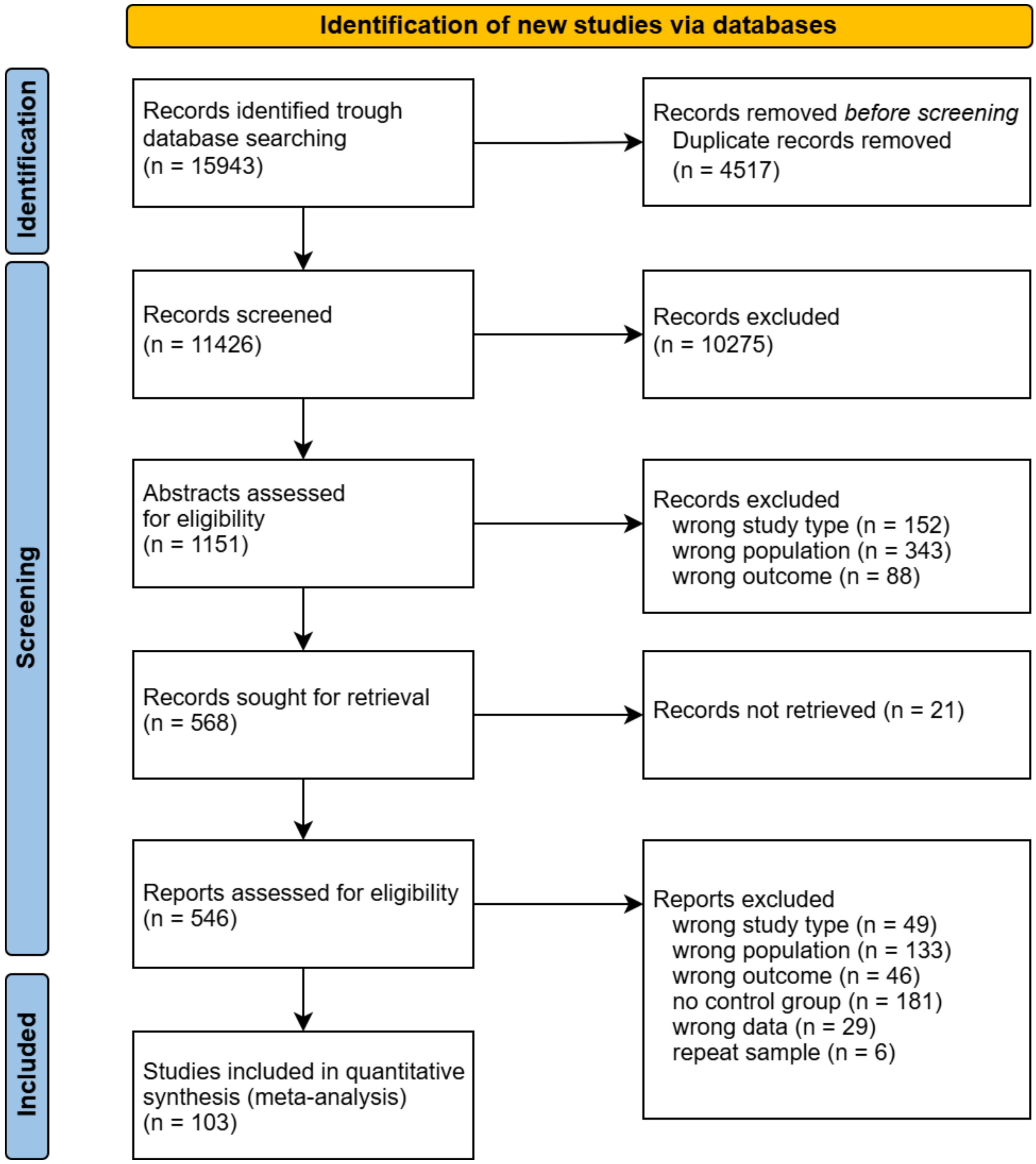
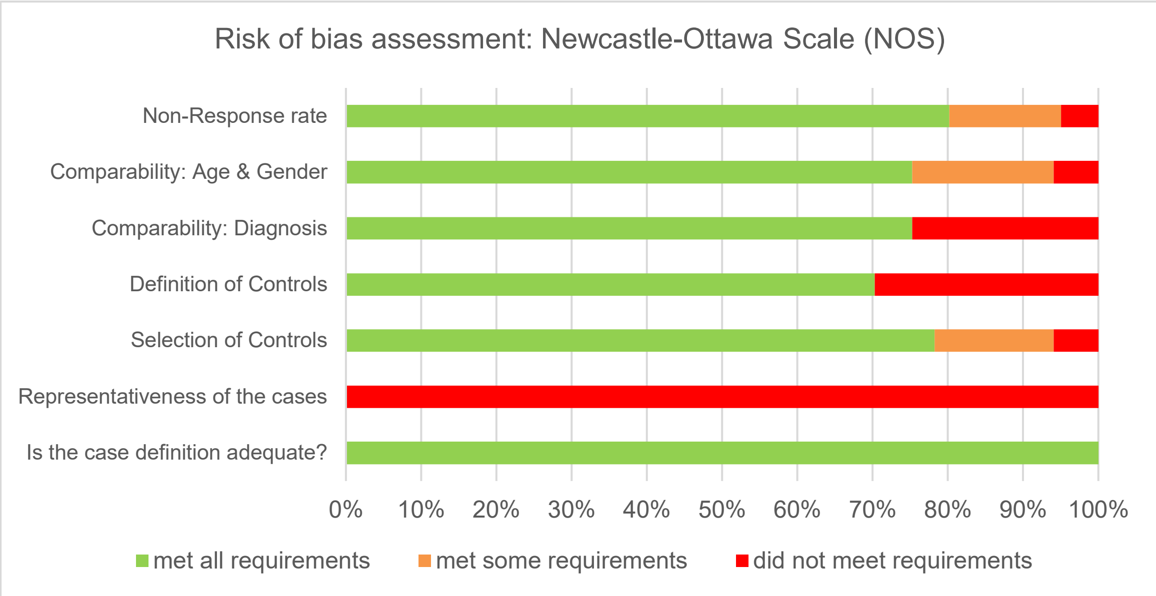


Figure 2

Risk of Bias assessment



This figure shows the results from the risk of bias assessment using the Newcastle Ottawa Scale. Green indicates the amount of studies which met criteria (rating A), orange indicates when studies partially met criteria (B), and red indicates when studies did not meet criteria (C).

Figure 3

Forest plot of the cognitive domains

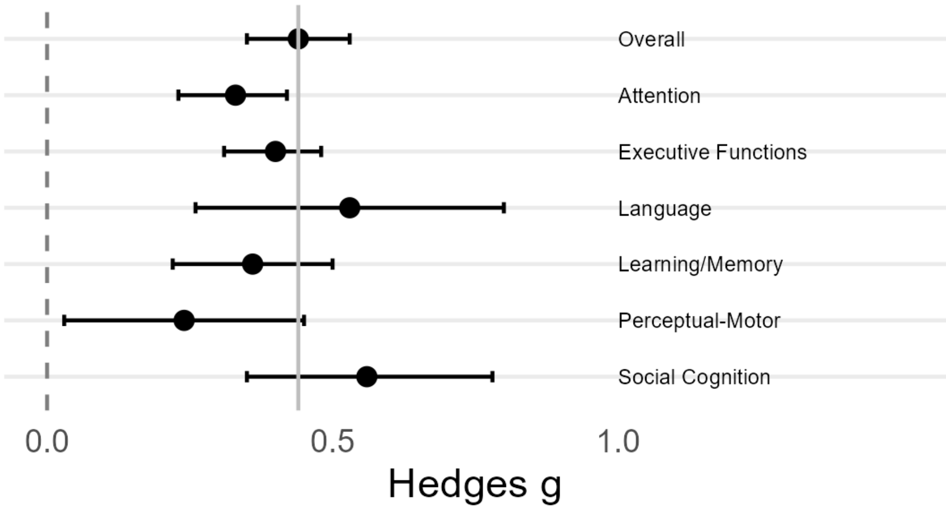


Figure 4a

Funnel plot of all effect sizes for all domains

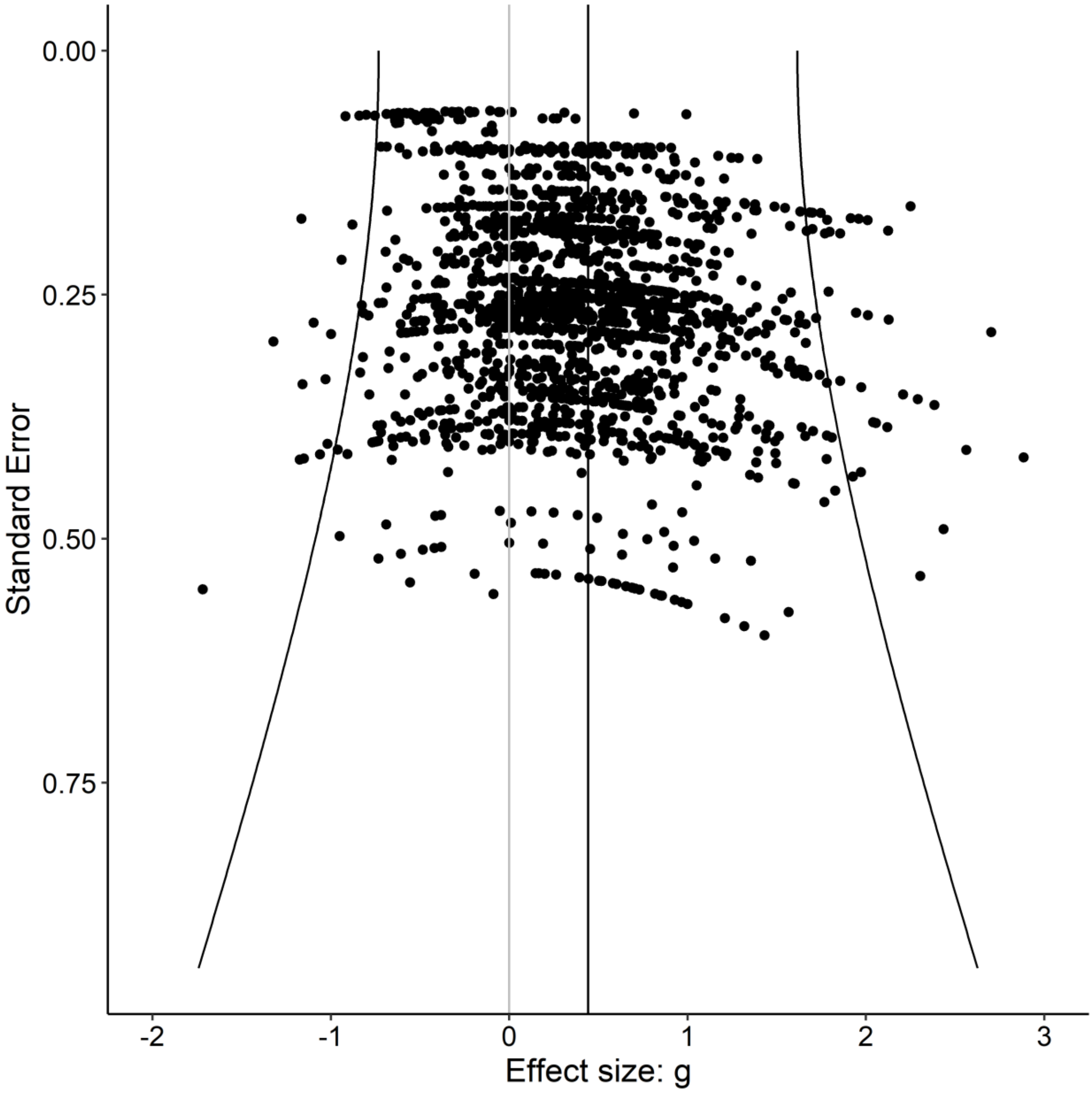
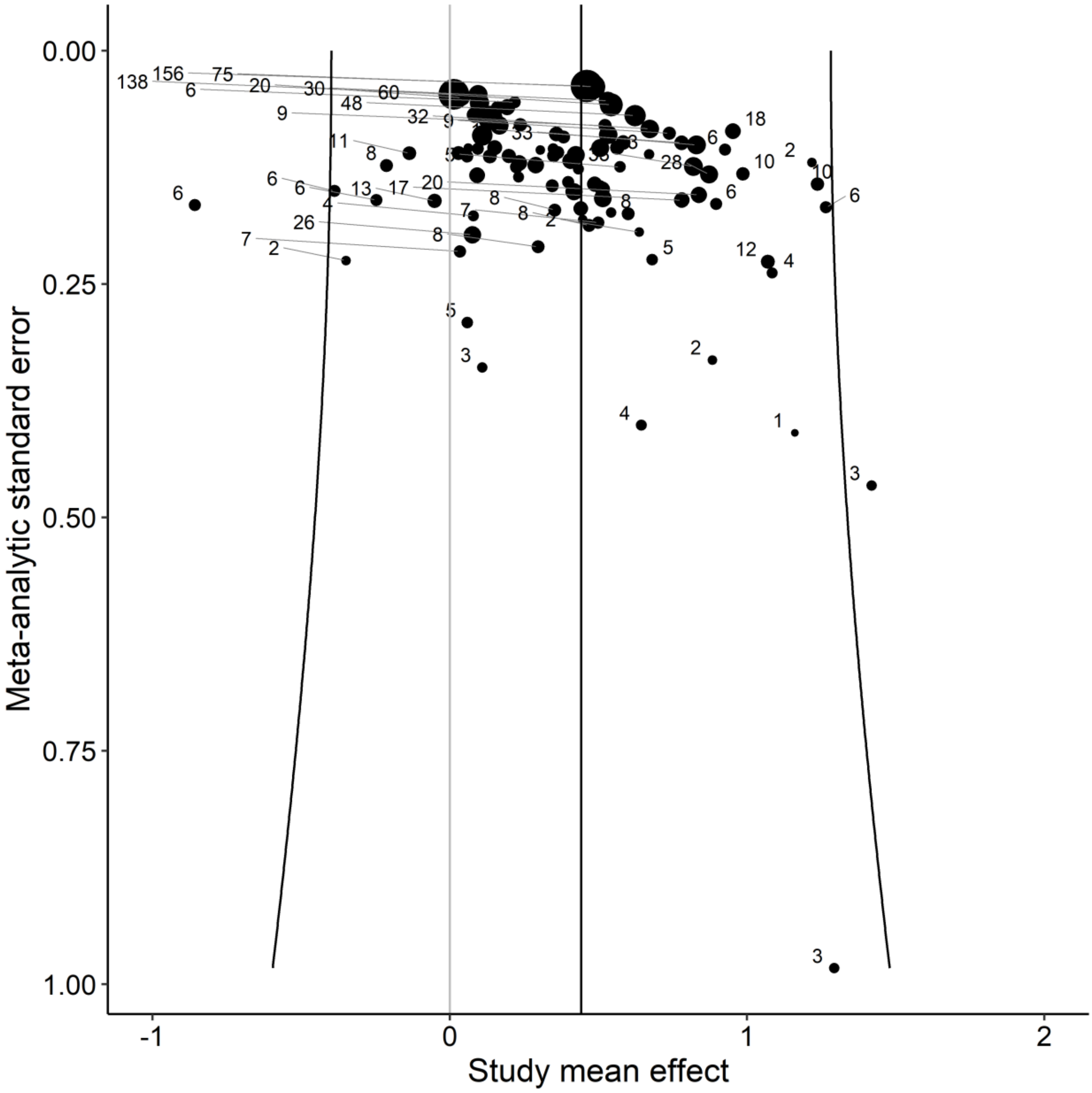


Figure 4b

Funnel plot of study effects for all domains.



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Tables and Figures

Table 1

Results for the Mean Effect Sizes for Differences Between Offenders and Comparisons per Domain

Domain	# studies	# tests	# ES	Mean g (SE)	95% CI	σ^2_{level2}	σ^2_{level3}	% var lvl 1	% var lvl 2	% var lvl 3	Q (df)
Overall	101	183	1756	0.44 (0.05)***	0.35, 0.52	0.19	0.16	8.73	49.34	41.93	17905.21 (1735)***
Attention	44	26	219	0.33 (0.05)***	0.23, 0.42	0.11	0.05	21.6	53.97	24.43	1091.09 (216)***
Executive Functions	74	75	812	0.40 (0.04)***	0.31, 0.48	0.17	0.10	14.35	54.63	31.02	5592.81 (794)***
Working Memory	32	20	90	0.44 (0.06)***	0.31, 0.56	0.13	0.04	18.12	63.90	17.99	576.12 (85)***
Inhibition	34	11	128	0.26 (0.06)***	0.15, 0.37	0.10	0.05	30.43	47.17	22.40	421.21 (122)***
Cognitive Flex.	40	11	252	0.51 (0.08)***	0.36, 0.66	0.09	0.20	12.12	26.10	61.78	2383.26 (251)***
Language	13	8	24	0.53 (0.13)***	0.26, 0.80	0.35	0.00	6.00	94.00	0.00	418.19 (23)***
Learning/Memory	24	19	152	0.36 (0.07)***	0.22, 0.50	0.16	0.04	15.74	66.29	17.97	1287.16 (149)***
Perceptual-Motor	17	21	94	0.24 (0.11)*	0.03, 0.45	0.19	0.07	2.99	68.45	28.56	2247.26 (93)***
Social Cognition	34	33	370	0.56 (0.11)***	0.35, 0.78	0.31	0.32	7.94	45.52	46.54	3199.47 (369)***

* $p < .05$ ** $p < .01$ *** $p < .001$.

Note. CI = confidence interval; ES = effect sizes; σ^2_{level2} = variance between effect sizes (within studies); σ^2_{level3} = variance between effect sizes (between studies); % var. = percentage of variance explained (analogous to an I^2 calculation for each level).

Table 2

Results of Moderator Analyses

Moderator	# Studies	# ES	β_0, Mean g (SE)	95% CI	β_1 (95% CI)	F(df_1, df_2)	p	σ^2_{level2}	σ^2_{level3}
Study Characteristics									
Publication year	100	1730	0.43 (0.04)***	0.35, 0.52	0.00 (−0.01, 0.02)	F(1, 1728) = 0.29	.589	.19***	.16***
Study Location						F(3, 1628) = 4.79	.003	.18***	.11***
Western Europe (RC)	41	589	0.42 (0.06)***	0.31, 0.53					
Europe: other	29	620	0.61 (0.07)***	0.47, 0.74	0.19 (0.01, 0.36)*				
Northern America	18	400	0.27 (0.09)**	0.10, 0.44	−0.15 (−0.35, 0.06)				
Other	7	41	0.11 (0.16)	−0.20, 0.41	−0.31 (−0.64, 0.02)				
Sample Characteristics									
Age	96	1650	0.43 (0.04)***	0.35, 0.51	-0.01 (-0.02, 0.00)	F(1, 1648) = 2.87	.090	.18***	.13***
% Male	100	1732	0.44 (0.04)***	0.35, 0.53	0.00 (−0.00, 0.00)	F(1, 1730) = 0.00	.991	.19***	.16***
% Minority group	23	408	0.53 (0.09)***	0.36, 0.71	−0.00 (−0.01, 0.00)	F(1, 406) = 0.58	.447	.13***	.15***
Groups matched for intelligence and/or education level						F(1, 1150) = 28.85	<.001	.16***	.18***
Yes (RC)	62	1066	0.36 (0.06)***	0.25, 0.48					
No	11	95	0.87 (0.10)***	0.67, 1.07	0.51 (0.32, 0.69)***				
Psychiatric diagnosis						F(1, 1202) = 52.75	<.001	.16***	.12***
Both groups general (RC)	21	458	0.57 (0.07)***	0.43, 0.71					
Both groups clinical	26	326	0.05 (0.06)	-0.07, 0.17	-0.52 (−0.68, −0.36)***				

Off. clinical, comp. general	41	442	0.56 (0.05)***	0.46, 0.67	-0.01 (-0.15, 0.14)				
Brain damage excluded						F(1, 983) = 0.02	.339	.20***	.16***
Yes (RC)	38	720	0.48 (0.07)***	0.34, 0.62					
No	5	272	0.45 (0.08)***	0.30, 0.60	–0.03 (–0.43, 0.37)				
Intellectual disabilities excluded						F(1, 345) = 0.03	.866	.14***	.10***
Yes (RC)	20	250	0.37 (0.09)***	0.20, 0.55					
No	6	97	0.41 (0.16)*	0.08, 0.73	0.03 (-0.34, 0.40)				

Offender group characteristics									
Type of offender						F(3, 1131) = 0.53	.662	.21***	.17***
General Violence (RC)	29	452	0.40 (0.08)***	0.24, 0.55					
Domestic Violence	12	349	0.53 (0.14)***	0.26, 0.80	0.13 (−0.18, 0.45)				
Sexual Offences	16	275	0.48 (0.09)***	0.30, 0.66	0.08 (−0.07, 0.24)				
Property	5	80	0.53 (0.14)***	0.26, 0.80	0.04 (−0.34, 0.42)				
Setting						F(2, 1406) = 0.38	.681	.20***	.17***
Prison (RC)	37	710	0.45 (0.07)***	0.31, 0.58					
Inpatient	26	360	0.51 (0.08)***	0.36, 0.67	0.07 (−0.11, 0.24)				
Outpatient	14	356	0.41 (0.12)***	0.18, 0.65	−0.03 (−0.30, 0.24)				
Diagnosis						F(3, 996) = 1.75	.155	.19***	.15***
No Diagnosis (RC)	22	465	0.52 (0.09)***	0.34, 0.70					
Neurodevelopmental	3	51	0.54 (0.09)***	0.37, 0.70	−0.07 (−0.57, 0.43)				
Schizophrenia/psychotic	19	299	0.32 (0.09)***	0.14, 0.51	0.02 (−0.23, 0.26)				
Personality Disorder	15	192	0.32 (0.09)***	0.14, 0.51	−0.20 (−0.45, 0.05)				
Test characteristics									
Domain						F(5, 1644) = 1.59	.161	0.19***	0.16***
Perceptual-Motor (RC)	17	94	0.31 (0.08)***	0.15, 0.46					
Attention	44	219	0.41 (0.06)***	0.30, 0.53	0.10 (−0.04, 0.25)				

Executive Functions	74	812	0.44 (0.05)***	0.34, 0.54	0.13 (0.00, 0.26)*
Language	13	24	0.61 (0.12)***	0.38, 0.83	0.30 (0.07, 0.53)*
Learning/Memory	24	152	0.44 (0.07)***	0.31, 0.58	0.13 (−0.02, 0.28)
Social Cognition	34	370	0.44 (0.07)***	0.31, 0.57	0.13 (−0.05, 0.32)

* $p < .05$ ** $p < .01$ *** $p < .001$.

Note. CI = confidence interval; ES = effect sizes; σ^2_{level2} = variance between effect sizes (within studies); σ^2_{level3} = variance between effect sizes (between studies); % var. = percentage of variance explained (analogous to an I^2 calculation for each level).

Table 3

Results for Publication Bias Analyses

Domain	# studies	# effect sizes	Funnel plot test	Egger test	Trim-and-fill analyses		
					Left	Right	# ES > cutoff
Overall	101	1756	z = -0.71, *p* = .480	z = 12.54, *p* < .001	0	10	yes
Complex Attention	43	217	z = 0.00, *p* = .703	z = 0.40, *p* = .692	10	0	yes
Executive Functions	74	812	z = 0.42, *p* = .598	z = 5.10, *p* < .001	0	2	no
Language	12	24	z = -1.47, *p* = .143	z = 2.04, *p* = .042	0	1	no
Learning/Memory	24	152	z = -2.71, *p* = .007	z = 2.41, *p* = .016	12	5	yes
Perceptual-Motor Functions	17	94	z = -1.78, *p* = .075	z = 2.21, *p* = .027	0	0	no
Social Cognition	34	370	z = -0.45, *p* = .651	z = 8.77, *p* < .001	0	14	yes

Note. Left = number of imputed effect sizes on the left side of the funnel plot distribution; Right = number of imputed effect sizes on the right side of the funnel plot distribution; #ES > cutoff = the number of imputed effect sizes on the left side in relation to the cutoff value of the estimator of the trim-and-fill method proposed by Fernández-Castilla et al. (2021). This estimate is based on the population effect size and power (number of effect sizes).

Graphical abstract

Highlights

- Offenders show weaker cognitive test performance than non-offending comparison groups (g = 0.44, small to medium effect).

- Significant differences found across all DSM-5 cognitive domains, with the largest effects in social cognition (g = 0.56) and language (g = 0.53).
- Differences between offenders and non-offending comparisons were minimal when clinical comparison groups were used.
- Findings suggest weaker test performance may reflect general clinical dysfunction rather than offender-specific deficits.